

MASTER RISK ORACLE EVIDENCE PACK

Consolidated evidence base for DeFi/RWA protocol, vault and asset risk scoring — Aave V3 Base, Chainlink oracle dependencies, GENIUS Act/RWA framework, x402/ERC-8004 infrastructure.

Review cutoff: **11 September 2026**. This document is a *foundation* for a separate reasoning model to design and red-team a risk-scoring methodology. It does not select weights, thresholds, loss probabilities, or a scoring formula. It builds on, and does not replace, primary-source-knowledge-base.md (referred to below as **the Foundation Document**), which remains authoritative for x402, ERC-8004, the enacted GENIUS Act statutory text, and Aave/Chainlink conceptual mechanics. Where this document repeats Foundation Document conclusions, it does so only to keep this pack self-contained for the next model; the Foundation Document's own footnotes [^F1]–[^F30] are preserved by reference in the Source Registry (Section 17) rather than re-derived.

A statute, a proposed rule, a GitHub file on a moving branch, and a deployed contract at a specific block are different kinds of evidence. Nothing in this document should be read as a live on-chain snapshot: every numeric figure is timestamped to its source, and every value that requires a fresh on-chain read is marked **[LIVE-QUERY REQUIRED]** with the exact contract and function needed to obtain it.

1. Scope and methodology

1.1 What this document adds to the Foundation Document

The Foundation Document established primary-source conceptual grounding for x402, ERC-8004, the enacted GENIUS Act, Aave V3 protocol mechanics, and Chainlink oracle mechanics, but explicitly flagged as missing: (a) actual Base deployment state, (b) the actual oracle graph and its history, (c) governance and historical parameter changes, (d) economic/security outcome evidence, (e) the legal implementation register beyond July–August 2026, and (f) instrument-specific RWA/stablecoin originals. This document's new research (Sections 7–13, 16) targets exactly those gaps, current to 11 September 2026. Sections 2–6 carry the Foundation Document's conclusions forward at summary level so this pack is self-contained; they are not a re-investigation and should not be read as superseding the Foundation Document's more detailed treatment.

1.2 Analytical rules preserved from the Foundation Document

- No scoring formula, feature weighting, or loss-probability estimate is proposed anywhere in this document.
- Observable fact, measurable risk signal, interpretation, and uncertainty are kept in distinct columns/paragraphs throughout.
- Regulatory compliance (GENIUS Act, state licensing, audits) is never treated as equivalent to low risk.
- Double counting, false precision, gameable metrics, and regime-dependent conclusions are flagged explicitly where identified.
- Missing evidence is marked **UNKNOWN / NOT VERIFIED**, never silently filled.

1.3 New methodological notes specific to this pack

- **Tooling boundary.** This research was conducted with web search and page retrieval only; no direct RPC/JSON-RPC calls, no Basescan API key, and no subgraph queries were executed. Every on-chain numeric parameter below (LTV, liquidation threshold, liquidation bonus, supply/borrow caps, debt ceilings, reserve factors) that is not independently reproducible from a dated governance document is therefore marked for live verification rather than stated as current fact. This is a deliberate evidentiary choice, not an oversight: stating a scraped or cached number as "the current LTV" would be false precision, since Aave's Risk Steward automation (Section 9.3) can change several of these parameters within governance-set bounds without a new forum post.
- **Moving branches.** The Aave address book (`[aave-address-book]`) is a continuously updated GitHub repository. Two forks/mirrors of it were found to disagree on two oracle addresses for Base (Section 8.5, Section 18 consistency check). All address-book data below is tagged with the exact retrieval date and repository path; a production system should pin an immutable commit hash, not `[main]`.
- **Search-engine cache dates.** Some figures below (e.g., Aave's own markets UI) were retrieved through cached search snippets rather than a live page render, because the Aave app is a client-rendered single-page application that returns no static content to a plain fetch. These are marked accordingly and should be treated as illustrative recent observations, not certified current values.

2. x402 payment architecture (carried forward from the Foundation Document)

The Foundation Document's review of the x402 whitepaper (6 May 2025) and current v2 specification stands unmodified by this research pass. Key carried-forward conclusions,

condensed:

- x402 defines payment **negotiation/settlement plumbing** (HTTP 402 signaling, `authorization`/`upfront`/`escrow` flows, EIP-3009/Permit2/ERC-7710 EVM transfer schemes) — it is a payment-rail specification, not a collateral-safety or asset-quality standard.
- A `settlement_pending` response is non-terminal; timeouts do not prove non-payment. Facilitator behavior (`/supported`) must be checked per deployment; it is not uniform.
- The Bazaar discovery extension has an unresolved core-vs-guide discrepancy about whether `EXTENSION-RESPONSES` are forwarded to buyers — this was unresolved as of the Foundation Document's review and is **not re-investigated here**.
- **Boundary preserved:** x402 adoption, payment volume, or Bazaar ranking measure usage/demand, not the financial safety of any underlying asset, and must not be conflated with asset risk.

No new x402 research was conducted in this pass; see the Foundation Document Section A and its footnotes ^[^F1]–^[^F5] for full detail.

3. ERC-8004 identity/discovery/reputation architecture (carried forward)

Carried forward without modification:

- ERC-8004 (Draft, created 13 August 2025) defines Identity, Reputation, and Validation registries for agents. Ownership of an agent identity is transferable; metadata/endpoints are mutable; `x402Support`, `active`, and `supportedTrust` are self-declared, not verified facts.
- Reputation feedback is signed, application-tagged, and Sybil-manipulable; a validator's 0–100 score is a specific validator's opinion, not a calibrated safety probability, and can change after issuance.
- **Boundary preserved:** payment/agent-service reliability metrics (from x402 or ERC-8004) must be kept analytically separate from collateral/protocol safety unless a specific, independently justified and validated relationship is established. None is established in this document.

No new ERC-8004 research was conducted in this pass; see the Foundation Document Section A and footnote ^[^F6].

4. GENIUS Act and regulatory/RWA framework

4.1 Enacted statute (carried forward, unmodified)

Public Law 119-27 (139 Stat. 419), approved 18 July 2025, remains the controlling primary text. The Foundation Document's statutory summary table (scope, issuers, reserves,

maturity/tokenization, reuse restrictions, redemption, disclosure/examination, prudential rules, AML/orders, yield prohibition, state/federal thresholds, custody, insolvency treatment, insurance status, foreign issuers, classification) is carried forward without modification. The drafting cross-reference issue in section 4(a)(3)(A) referencing (1)(D) instead of the actual (1)(C) reserve-disclosure provision remains unresolved and should not be silently corrected.

4.2 Implementation status as of 11 September 2026 — NEW

The Foundation Document's review (dated through 19 August 2026) captured only the Treasury section 3 NPRM and an OCC news release trailer. This pass extends the regulatory register:

Regulator	Instrument	Date	Status as of 11 Sep 2026	Scope
Treasury (Dept. of the Treasury)	Advance Notice of Proposed Rulemaking, 90 FR 45159	19 September 2025	Closed; informed later NPRM	Solicited comment on safe harbors, statutory-term clarity, illicit-finance detection, foreign- regime comparability, tax/insurance implications
OCC	NPRM, new 12 CFR Part 15 (plus conforming amendments to Parts 3, 6, 8, 19), Docket OCC-2025-0372	Released 25 Feb 2026; published Federal Register 2 Mar 2026	Proposed rule only — not final. Comment period closed 1 May 2026. No final rule published as of this review.	Licensing, permissible activities, reserve- asset interpretation, anti-yield enforcement, diversification standards, redemption, custody, capital/liquidity, for national banks/subsidiaries, Federal-qualified nonbank issuers (FQPSIs), Foreign PSIs, and State-qualified issuers under OCC authority. ~376 pages, 211 public-comment questions.
FDIC	NPRM, RIN 3064- AG19	10 April 2026 (Federal Register)	Proposed rule only — not final. Comments closed 9 June 2026.	Prudential standards for FDIC-supervised IDI subsidiaries issuing payment stablecoins; deposit- insurance treatment of reserve deposits; tokenized-deposit

Regulator	Instrument	Date	Status as of 11 Sep 2026	Scope
FinCEN / OFAC (Treasury)	Joint NPRM on BSA/AML/sanctions compliance	10 April 2026	Proposed rule only — not final. Comments closed 9 June 2026.	treatment. Closely mirrors, and cross-references, the OCC proposal. AML program, sanctions-compliance program obligations for PPSIs, separate from the OCC's core prudential rule by design (statute splits this out).
Treasury	NPRM implementing Section 3 (issuance/offer/sale prohibitions, cross-border reach), Docket TREAS-DO-2026-0496, Federal Register Doc. 2026-16796	Published 18 August 2026	Proposed rule only — not final. Comment deadline 19 October 2026.	Defines "issue," "offer," "sell" in the U.S.; safe harbors; cross-border reach of the foreign-issuer restriction.
Federal Reserve Board	GENIUS Act implementing rule	—	UNKNOWN / NOT VERIFIED — no Federal Reserve NPRM was identified in this research pass. This is a gap, not a confirmation of absence; the Fed is a named primary regulator under the Act and its rule status should be checked directly at federalreserve.gov	—

Regulator	Instrument	Date	Status as of 11 Sep 2026	Scope
			before relying on this pack.	
NCUA	GENIUS Act implementing rule (credit-union issuers)	— (proposal referenced by secondary sources, not independently retrieved in full)	Secondary sources describe an NCUA NPRM paralleling FDIC's; primary NCUA document not independently fetched in this pass — mark NOT VERIFIED for specifics.	Credit-union-supervised issuers
State regulators	Substantial-similarity determinations, \$10bn threshold transition rules	—	UNKNOWN / NOT VERIFIED — no comprehensive state-by-state register was compiled in this pass.	State-qualified issuer framework

Key interpretive facts, none of which are in the Foundation Document:

- As of 11 September 2026, **no primary federal payment-stablecoin regulator has issued a final implementing rule.** All identified instruments are NPRMs (proposed rules) with open or recently-closed comment periods. This means the GENIUS Act's earlier effective-date trigger (120 days after final rules) has **not** been activated by any confirmed final rule; the statute's fallback effective date of **18 January 2027** (18 months after enactment) remains the only confirmed effective date as of this review, pending any final rule this pack has not identified.
- No entity has yet been publicly confirmed by this research as an approved "Federal qualified payment stablecoin issuer," "State qualified payment stablecoin issuer," or OCC-approved IDI-subsidiary PPSI.** Approvals cannot occur, by statutory design, before implementing rules and application processes are finalized. Any marketing claim that a specific token is "GENIUS Act approved," "GENIUS Act compliant," or a "permitted payment stablecoin issuer" as of 11 September 2026 should be treated as **unverified and premature** pending confirmation from the relevant regulator's own issuer registry (not yet established).
- A secondary trade-press source (PYMNTS, 11 September 2026, citing a 1 September 2026 Braumiller Law Group overview) states that prospective issuers are already redesigning

compliance systems in anticipation of the rules, including wallet-level monitoring and preparation for machine-agent-initiated payments — this is a market-behavior observation, not a legal status confirmation, and is included here only as context for how close the industry believes finalization is, not as evidence of any current legal status.

- **This has direct relevance to Aave Base RWA/stablecoin scoring:** none of the stablecoins currently listed as Aave V3 Base reserves (USDC, USDbC, EURC, GHO — Section 7) currently operate, or can currently operate, under a GENIUS Act PPSI license, because no such license yet exists. Their present legal basis is pre-GENIUS-Act state money-transmission/trust-company frameworks (for Circle's USDC/EURC) or Aave-DAO-governed on-chain over-collateralization (for GHO — Section 11.3), not the GENIUS Act framework. A risk methodology must not credit any Base reserve asset with "GENIUS Act compliance" today.

4.3 What remains genuinely unresolved from the Foundation Document

The Foundation Document's outstanding items (comprehensive dated regulator/Federal Register check, section 20 trigger determination, foreign-comparability/registration lists, state substantial-similarity determinations) remain outstanding after this pass. This pass narrows but does not close those gaps — see Section 16.

5. Aave V3 protocol mechanics (carried forward, with new additions)

5.1 Carried forward from the Foundation Document, unmodified

- Health factor = liquidation-threshold-weighted collateral value ÷ debt value, computed by `GenericLogic` using live oracle prices, token precision, enabled-collateral flags, and effective eMode; no-debt sentinel is `uint256.max`.
- Liquidation validation requires HF strictly below 1 plus reserve/user/debt/grace-period conditions; the pinned `LiquidationLogic` 50% close-factor restriction applies only above the \$2,000-equivalent collateral-and-debt thresholds and HF > 0.95; below that, full debt can be liquidated subject to constraints; \$1,000-equivalent dust limits apply when both sides remain non-zero. These are revision-specific constants, not universal.
- V3.3 introduced position-wide close-factor treatment and reserve-deficit accounting for bad debt (burning debt tokens recognizes a deficit; it does not recover assets).
- Isolation mode (legacy): one isolated collateral asset, restricted borrow assets, a debt ceiling. V3.6 decoupled non-default eMode permissions from ordinary reserve settings. V3.7 removes legacy isolation and "borrowable in isolation," and introduces isolated eMode instead (Section 5.3).

- Interest behavior is strategy-dependent; zero caps mean "uncapped" in documented configuration, not "zero activity."

5.2 New: confirmed Aave protocol version on Base

Aave V3 Base is on protocol version 3.7 as of this review. Evidence chain:

- Aave's official changelog (aave.com/docs/resources/changelog, primary source) states: "*Aave v3.7 completes its rollout with deployments to Ethereum (Core, Lido), Polygon, Avalanche, Arbitrum, Base, BNB Chain, Linea, Plasma, and Mantle,*" with the v3.6 entry separately dated 9 January 2026.
- A governance execution record (vote.onaave.com/governance.aave.com, cross-referenced via search cache), "*Upgrade Aave instances to v3.7 Part 1,*" lists Part 1 networks as Sonic, Optimism, Gnosis, Scroll, Celo, MegaEth, XLayer, and Ethereum (EtherFi) — **Base is not in Part 1**, implying Base was upgraded in the unretrieved "Part 2" execution, consistent with the changelog's later "rollout completes" framing.
- Independent secondary sourcing (CoinMarketCap CMC AI page, dated 27 June 2026) states v3.7 rollout completion on **29 May 2026** across ten networks — this date is **secondary, not independently confirmed from a primary governance-execution timestamp for the Base-specific payload in this pass**, and should be verified against the actual on-chain execution transaction for the Base [PoolConfigurator](#) implementation upgrade before being treated as exact.
- **Practical consequence for oracle-dependency work:** Aave's own v3.7 feature notes (ARFC "BGD. Aave v3.7," governance.aave.com, primary) state v3.7 **removes the L2 sequencer oracle** (i.e., removes the [PriceOracleSentinel](#)/sequencer-uptime gating that used to delay liquidations after a sequencer outage) and makes the reserves list append-only, in addition to introducing the eMode isolated flag. This directly updates the Foundation Document's oracle-notes finding (its footnote [^{F19}]) that "V3.7 also removes prior sequencer-sentinel gating" — this is now confirmed as the *actual* Base-deployed behavior, not merely a documented future-branch feature. See Section 10.5 for why this matters given Base's actual 2025–2026 sequencer-outage history.

5.3 New: Aave V3.7 isolated eMode mechanism (Base-relevant)

Per the same ARFC, v3.7's isolated eMode design allows some already-enabled, out-of-eMode-category collateral to retain liquidation-threshold contribution while carrying zero LTV for new borrowing within that eMode — this generalizes what the Foundation Document described from the source-branch review (its footnote [^{F18}]) into a now-confirmed live-network behavior. An **active governance discussion is underway as of this review**: ARFC "Unified Handling of the Isolated Flag in Aave V3.7 E-Modes" (governance.aave.com, dated in this review's window) proposes a single rule for when an eMode category should be flagged "isolated," recommending isolation for most wstETH/ETHx-looping categories plus WETH/WPOL-correlated categories on MegaETH, Gnosis, and Polygon. **This proposal's text**

as retrieved did not confirm or deny which of Base's 15 eMode categories (Section 7.4) are currently flagged isolated — this is a live parameter requiring on-chain verification (`Pool.getEModeCategoryCollateralConfig` / the v3.7 eMode struct's isolation flag) before it can be used as a risk input. Marked **[LIVE-QUERY REQUIRED]**.

5.4 New: Umbrella backstop (replaces the Safety Module referenced by general Aave literature)

Secondary reporting on the April 2026 incident (Section 10.1) states Aave **"launched its Umbrella backstop system in late 2025, replacing the older Safety Module."** Under Umbrella, holders can stake yield-bearing tokens (e.g., aWETH) into per-asset insurance vaults that can be slashed by DAO vote to cover a recognized deficit/bad debt event. This is a materially different backstop architecture from the classic AAVE-staked Safety Module described in older Aave literature, and from the "Safety Module" language that may appear in older secondary sources. **This finding is from secondary reporting only in this pass (Forbes, 18 April 2026) and was not independently cross-checked against an Aave-published Umbrella specification document; treat the existence and mechanics of Umbrella as probable but not fully primary-verified, and verify against `aave.com/docs` or the Umbrella contracts repository before relying on it.** If confirmed, "Umbrella vault stake and slashing history" is a distinct candidate solvency-backstop signal, separate from protocol TVL or reserve factors (see Section 14).

5.5 New: Risk Steward / automated parameter-update layer

The Foundation Document treated caps, freezes, and parameter changes as governance-controlled facts without describing the mechanism by which many of them are actually set. This pass found that a substantial share of Aave's ordinary (non-emergency) risk-parameter changes are **not** individually voted on-chain by AAVE holders; they are executed through a constrained-automation layer:

- **Risk Steward contracts** (BGD Labs, `aave-dao/aave-v3-risk-stewards`, primary GitHub repo): smart contracts holding a governance-delegated, **bounded** permission to adjust specific parameters (rate-strategy curves, supply/borrow caps, LTV/LT/LB/debt-ceiling/eMode-category assignment, and LST/stable price-cap parameters) without a full governance vote, subject to hard-coded bound checks (e.g., historical cap increases limited to roughly 2x the current configuration per adjustment).
- **Edge Risk Oracle** (Chaos Labs, primary blog post "Introducing Aave's Chaos Labs Risk Oracles"): an off-chain analytics engine that monitors utilization, cap thresholds, volatility, and anomalous behavior, and pushes recommended parameter updates on-chain via a Chainlink-Automation-compatible `AaveStewardsInjector` middleware contract. Injected updates expire after a **6-hour window** if not applied.

- **Base-specific deployed instances** (from the address book, Section 7.2): `RISK_STEWARD`, `RISK_COUNCIL`, `EDGE_RISK_STEWARD_CAPS`, `EDGE_INJECTOR_CAPS`, and `EDGE_RISK_ORACLE` all have distinct deployed addresses on Base, confirming this automation layer is live on the Base deployment, not just on Ethereum mainnet.
- **Risk-methodology implication:** because caps, rates, and some collateral parameters can change automatically within bounds multiple times without a discrete governance forum post, a risk oracle that only ingests "governance proposal" events as its change-history source will under-count actual parameter change frequency. The raw on-chain parameter-change event log (not just forum/AIP history) is the more complete primary source for change-frequency signals.
- Two further Base-deployed contracts, `CLINIC_STEWARD` and `POOL_EXPOSURE_STEWARD`, appear in the address book with names suggesting bad-debt/dust-position cleanup and cross-asset exposure-limiting functions respectively, consistent with the general "dust position" and "deficit" problems documented in Aave's own v3.3 feature notes (already cited in the Foundation Document, footnote [^F16]). **Their exact function and permission scope were not independently confirmed against a contract-level specification in this pass — mark UNKNOWN / NOT VERIFIED beyond the inference from naming convention and adjacent documentation.**

6. Chainlink oracle architecture (carried forward, with new additions)

6.1 Carried forward from the Foundation Document, unmodified

- Path: Aave reserve → `AaveOracle` → adapter → one or more price/ratio providers. `AaveOracle` accepts positive `latestAnswer()` values; missing/non-positive triggers fallback; positive-but-stale values do **not** trigger fallback; an upstream revert is not caught.
- Aave's stable adapter takes the lower of the upstream answer and a configured upper cap (does not floor a depeg at \$1). Ratio-based (LST/LRT) adapters bound growth using a snapshot ratio, elapsed time, and a maximum configured yearly growth rate — a capped ratio is a transformation, not proof of an attack, and a configured fallback may not function as expected.
- Heartbeat/deviation trigger updates; they do not guarantee continuous streaming. `answeredInRound` is documented as deprecated and should not be used as a standalone freshness test.
- Chainlink separates market-integrity risk (thin markets, correlated node/source failure) from application-code risk; sourcing methodology differs across multi-vendor market prices, authoritative NAV/reserve publishers, and on-chain exchange rates.

- L2 sequencer uptime feeds exist for Base (0 = up, 1 = down, with `startedAt`); recovery-grace logic belongs to the consumer, not the feed. **This is now superseded in practice for Aave V3 Base specifically by the V3.7 sequencer-sentinel removal (Section 5.2) — the feed may still exist and be queryable on Base, but Aave’s own Pool contract no longer consults it for liquidation gating as of the v3.7 upgrade.**

6.2 New: Aave’s custom price-feed adapter taxonomy (`aave-price-feeds` repository, primary)

The Foundation Document referenced `PriceCapAdapterStable.sol` and `PriceCapAdapterBase.sol` from a pinned revision. This pass retrieved the repository’s own README, which describes three adapter families used across all Aave deployments (Base included, subject to per-asset confirmation in Section 8):

Adapter family	Purpose	Mechanism
<code>PriceCapAdapterBase</code> (rate-based cap)	LSTs/LRTs that grow in value relative to an underlying (wstETH, rsETH, weETH, ezETH, osETH, and structurally similar assets)	Stores a snapshot ratio + timestamp; computes $\text{maxRatio} = \text{snapshotRatio} + \text{maxGrowthPerSecond} \times \text{elapsed}$; returns the capped ratio if the live ratio exceeds it. Documented design goal is "sufficient margin," not maximum precision — Aave’s own documentation states other risk parameters (LTV/LT) are expected to provide additional protection, i.e., the price adapter is not meant to be the sole safeguard.
<code>PriceCapAdapterStable</code> (fixed cap)	USD-pegged stablecoins (USDC, USDT, DAI, EURC, RLUSD, and similar)	Caps the reported price at a fixed configured maximum (e.g., \$1.04 for a 4% cap); returns the capped price if the live price exceeds it. Does not floor a depeg — a stablecoin trading below the cap is passed through at its live (lower) price.
Composed/derived feeds	Non-direct pairs (e.g., asset/ETH × ETH/USD → asset/USD)	Combines two upstream feeds; the repository’s own documentation states composition itself is not capped — only the base rate-cap or stable-cap adapters cap their output.

A separate, documented, **asset-specific precedent** is directly relevant to Base’s tBTC/cbBTC/LBTC reserves: a 17 July 2025 governance thread ("[Direct to AIP] LBTC Oracle and CAPO implementation Update," governance.aave.com, primary) records that Chaos Labs migrated Ethereum’s LBTC price source **from a flat BTC/USD Chainlink feed to a composite**

LBTC-BTC exchange-rate feed wrapped in a CAPO limiter, specifically because Lombard began converting LBTC staking rewards into additional LBTC (a yield-bearing redesign effective 22 July 2025), which would otherwise cause Aave to under-price LBTC's redeemable value if it continued using a flat 1:1 BTC assumption. This establishes that Aave's own risk team treats "flat peg-asset feed" vs. "ratio-aware ISO feed" as a materially different risk posture for exactly this asset class — which is why the Base address-book finding in Section 8.5 (tBTC and cbBTC sharing one oracle address) is flagged as requiring verification rather than assumed benign.

6.3 New: dedicated Aave/SVR proxy feeds

Chainlink's individual feed pages (data.chain.link, primary, spot-checked for ETH/USD, BTC/USD, USDC/USD, USDT/USD, LINK/USD on Ethereum mainnet) each list, alongside the "Standard" proxy address and an "SVR Proxy" (Smart Value Recapture, for OEV/MEV recapture), a **separate "AAVE SVR Proxy" address** — i.e., Aave has its own dedicated proxy contracts for at least some major feeds, distinct from the general-purpose SVR proxy used by other integrators. **Whether any Base-deployed Aave reserve currently points at an AAVE-dedicated SVR proxy, a generic SVR proxy, or a plain standard proxy was not determined for any specific Base asset in this pass** — this is a per-asset, per-chain fact requiring the live-query method in Section 12, and matters because SVR proxies have different update/incentive dynamics (they exist specifically to change who captures value from oracle-triggered liquidations) than a standard proxy.

6.4 Base's non-Chainlink oracle landscape (context, not an Aave-Base finding)

Base's own documentation (docs.base.org/chain/oracles, primary) lists Chainlink, API3, Chronicle, DIA, Pyth, and Gelato/RedStone-style providers as available on Base generally. **This is general Base-chain infrastructure, not evidence about which oracle any specific Aave Base reserve actually uses** — Aave's `AaveOracle` on Base is confirmed (address book) to route through Aave-governed adapter addresses per asset (Section 8), and nothing in this research suggests Aave Base consumes non-Chainlink sources directly, but this was not exhaustively ruled out for every asset.

7. Aave V3 Base deployment evidence

7.1 Deployment identity

Field	Value	Source
Chain	Base Mainnet, chain ID 8453	Address book (<code>AaveV3Base.ts</code> , <code>CHAIN_ID = 8453</code>)
Protocol version	Aave V3.7 (confirmed — see Section 5.2)	Aave changelog + governance ARFC <code>aave-dao/aave-</code> <code>address-book</code> , <code>AaveV3Base.sol</code> , retrieved 11 Sep 2026
<code>POOL_ADDRESSES_PROVIDER</code>	<code>0xe20fCBdBfFC4Dd138cE8b2E6FBb6CB49777ad64D</code>	same
<code>POOL</code> (proxy)	<code>0xA238Dd80C259a72e81d7e4664a9801593F98d1c5</code>	same
<code>POOL_CONFIGURATOR</code> (proxy)	<code>0x5731a04B1E775f0fdd454Bf70f3335886e9A96be</code>	same
<code>ORACLE</code> (<code>AaveOracle</code>)	<code>0x2Cc0Fc26eD4563A5ce5e8bdcfe1A2878676Ae156</code>	same
<code>ACL_MANAGER</code>	<code>0x43955b0899Ab7232E3a454cf84AedD22Ad46FD33</code>	same
<code>ACL_ADMIN</code>	<code>0x9390B1735def18560c509E2d0bc090E9d6BA257a</code>	same
<code>AAVE_PROTOCOL_DATA_PROVIDER</code> (Pool Data Provider)	<code>0x0F43731EB8d45A581f4a36DD74F5f358bc90C73A</code>	same
<code>POOL_IMPL</code>	<code>0xA4AbC5FcBA6D0d7E3D144d6dbF6cb6128599dFdB</code>	same
<code>POOL_CONFIGURATOR_IMPL</code>	<code>0xFF42Ce30054dcE7dc7C1282a9a497aa58eaBCe99</code>	same
<code>COLLECTOR</code> (treasury)	<code>0xBA9424d650A4F5c80a0dA641254d1AcCE2A37057</code>	same
<code>DEFAULT_INCENTIVES_CONTROLLER</code>	<code>0xf9cc4F0D883F1a1eb2c253bdb46c254Ca51E1F44</code>	same
<code>EMISSION_MANAGER</code>	<code>0x6533A273F3aC84Df91DCD654D6EBAAbA73687e246</code>	same

Field	Value	Source
DEFAULT_A_TOKEN_IMPL	0x273E4B97C3f5280AFf4949Aa19a27fF54968458D	same
DEFAULT_VARIABLE_DEBT_TOKEN_IMPL	0x7354Dc700A1A2ab9622f2292B60Ca1ceD5B204D0	same
RISK_STEWARD	0x494bcfd3937aBdebEF3D2c2eae1CE8A2Fb629032	same
RISK_COUNCIL	0xfbeB4AcB31340bA4de9C87B11dfBf7e2bc8C0bF1	same
EDGE_RISK_STEWARD_CAPS	0xB892202d9Ce2C16C565A492a5168689b215Eb269	same
EDGE_INJECTOR_CAPS	0x4f84A364B66Eb6280350da011829a6BD02B4712f	same
EDGE_RISK_ORACLE	0x239d3Bc5fa247337287cb03f53B8bc63DBBc332D	same
CLINIC_STEWARD	0x1a79098C3149020ACdEfb3EE44AA1d9509305530	same
POOL_EXPOSURE_STEWARD (variant A)	0x4f1F3E32f20847a4c9F002882eB607aD750b6115	aave-dao/aave-address-book , retrieved 11 Sep 2026
POOL_EXPOSURE_STEWARD (variant B — conflicts with variant A)	0x4f1F3E32f20847a4c9F002882eB607aD750b6115 (ts export) vs. distinct listing seen in a bgd-labs fork snapshot	See Section 18 consistency check
DEBT_SWAP_ADAPTER	0xb12e82DF057BF16ecFa89D7D089dc7E5C1Dc057B	aave-dao/aave-address-book
WITHDRAW_SWAP_ADAPTER	0x5598BbFA2f4fE8151f45bBA0a3edE1b54B51a0a9	same
SWAP_COLLATERAL_ADAPTER	0x2E549104c516b8657A7D888494DfbAbD7C70b464	same
REPAY_WITH_COLLATERAL_ADAPTER	0x63dfa7c09Dc2Ff4030d6B8Dc2ce6262BF898C8A4	same
CONFIG_ENGINE	0xa5008BC7Fd697F458482205Ed52Dc51e91d37446	same
L2_ENCODER	0x39e97c588B2907Fb67F44fea256Ae3BA064207C5	same
UI_POOL_DATA_PROVIDER	0x0C6BC4a12039788be08F87e87Cff87FEDbd1D386	same
UI_INCENTIVE_DATA_PROVIDER	0x91E04cf78e53aEB609e8a7f2003e7EECD743F2B	same

Field	Value	Source
WALLET_BALANCE_PROVIDER	0x5779b29B0a34577d927E8D511B595ef9abbFAE82	same
WETH_GATEWAY	0xa0d9C1E9E48Ca30c8d8C3B5D69FF5dc1f6DFfC24	same
STATA_FACTORY	0x78d33BF0014ab169725F2Ea5a62b200F2977faeE	same
DUST_BIN	0x3870D459d2fC2Edbb03E0DdeE6c0Ccb268caf202	same
Governance <code>Executor</code> /robot on Base	0x2DC219E716793fb4b21548C0f009Ba3Af753ab01 (Payloads Controller / Executor)	aave-dao/aave-governance-v3 README
Governance short-timelock admin address on Base	0x9390B1735def18560c509E2d0bc090E9d6BA257a (matches <code>ACL_ADMIN</code> above)	same

Retrieval provenance: the full `AaveV3Base.sol` and `AaveV3BaseAssets`/`AaveV3BaseEModes`/`AaveV3BaseExternalLibraries` libraries were retrieved by direct fetch of `github.com/aave-dao/aave-address-book/blob/main/src/AaveV3Base.sol` on 11 September 2026, HEAD of `main` (503 lines / 24.6 KB at retrieval time, no specific commit SHA captured — **this is a moving-branch citation and should be re-pinned to a commit hash before production use**).

7.2 Reserve/asset universe (15 listed assets)

Symbol	Underlying address	Decimals	aToken
WETH	0x4200	18	0xD4a0e0b9149BCee3C920d2E00b5dE
cbETH	0x2Ae3F1Ec7F1F5012CFEab0185bfc7aa3cf0DEc22	18	0xcfc3D55c10DB69f28fD1A75Bd73f3C
USDbC (bridged, legacy)	0xd9aAEc86B65D86f6A7B5B1b0c42FFA531710b6CA	6	0x0a1d576f3eFeF75b330424287a95A
wstETH	0xc1CBa3fCea344f92D9239c08C0568f6F2F0ee452	18	0x99CBC45ea5bb7eF3a5BC08FB1B7E5

Symbol	Underlying address	Decimals	aToken
USDC (native Circle)	<code>0x833589fCD6eDb6E08f4c7C32D4f71b54bdA02913</code>	6	<code>0x4e65fE4DbA92790696d040ac24Aa4</code>
weETH	<code>0x04C0599Ae5A44757c0af6F9eC3b93da8976c150A</code>	18	<code>0x7C307e128efA31F540F2E2d976C99</code>
cbBTC	<code>0xcbB7C0000aB88B473b1f5aFd9ef808440eed33Bf</code>	8	<code>0xBdb9300b7CDE636d9cD4AFF00f6F6</code>
ezETH	<code>0x2416092f143378750bb29b79eD961ab195CcEea5</code>	18	<code>0xDD5745756C2de109183c6B5bB886F</code>
GHO	<code>0x6Bb7a212910682DCFdbd5BCBb3e28FB4E8da10Ee</code>	18	<code>0x067ae75628177FD257c2B1e500993</code>
wrsETH	<code>0xEDfa23602D0EC14714057867A78d01e94176BEA0</code>	18	<code>0x80a94C36747CF51b2FbabDfF045f6</code>
LBTC	<code>0xecAc9C5F704e954931349Da37F60E39f515c11c1</code>	8	<code>0x90072A4aA69B5Eb74984Ab823EFC5</code>
EURC	<code>0x60a3E35Cc302bFA44Cb288Bc5a4F316Fdb1adb42</code>	6	<code>0x90DA57E0A6C0d166Bf15764E03b83</code>
AAVE	<code>0x63706e401c06ac8513145b7687A14804d17f814b</code>	18	<code>0x67EAF2BeE4384a2f84Da9Eb8105C6</code>
tBTC	<code>0x236aa50979D5f3De3Bd1Eeb40E81137F22ab794b</code>	18	<code>0xbccFFB4B3beADc989Bd1458740952a</code>
syrupUSDC	<code>0x660975730059246A68521a3e2FBD4740173100f5</code>	6	<code>0xD7424238CcbE7b7198Ab3cFE232e6</code>

Additionally, every asset has an associated `INTEREST_RATE_STRATEGY` address; notably, **12 of the 15 assets (all except WETH/cbETH/USDbC's group being also shared, GHO, and syrupUSDC's own listing) share the identical interest-rate-strategy contract address** `0x86AB1C62A8bf868E1b3E1ab87d587Aba6fbCbDC5`. This is expected under Aave's V3 default interest-rate-strategy design (one strategy contract can serve many reserves with reserve-specific parameters passed at call time) but is worth recording as an observable fact: it means "which contract is called" is not a useful differentiator between these assets' rate behavior — the actual optimal-utilization/slope parameters (not captured in the address book) are what differ, and those require a live read.

Several assets also have `STATIC_A_TOKEN` and/or `STATA_TOKEN` wrapper addresses (ERC-4626-style wrapped aToken vaults) — WETH, cbETH, USDbC, wstETH, USDC, weETH, cbBTC, GHO, wrsETH, AAVE, EURC, and syrupUSDC all list a `STATA_TOKEN`; ezETH and tBTC do not list one in the retrieved file, and LBTC lists neither. **This is a candidate double-counting hazard:** aToken

balances and their corresponding StataToken-wrapped balances represent the same underlying claim through two different token contracts; a data pipeline that sums raw token balances across both without deduplication will overstate exposure.

7.3 Assets not currently listed on Aave V3 Base (explicitly absent)

Based on the retrieved asset list, Base does **not** list (as of retrieval): DAI, USDT, WBTC (unwrapped), rETH, osETH, sUSDe/USDe, RLUSD, or any Pendle Principal Tokens (PTs), even though several of these appear on other Aave V3 deployments (Ethereum, Arbitrum, X Layer, Monad) per governance-forum listings found in this research (Section 9). Absence of a listing is itself a risk-relevant fact (it bounds the asset universe this pack needs to score) and should not be inferred to mean anything about those assets' quality — Aave lists assets based on demand, deployment-specific governance decisions, and risk-service-provider recommendations, not a uniform cross-chain policy.

7.4 eMode categories (15 defined, IDs 0–15, ID 0 = "none")

ID	Name (as encoded in address book)	Interpretation (not independently verified against live parameters)
0	<code>NONE</code>	Default / no eMode
1	<code>WETH_cbETH_wstETH_weETH_WETH</code>	ETH-correlated collateral basket, borrowable: WETH
2	<code>ezETH_wstETH</code>	ezETH collateral, borrowable: wstETH
3	<code>ezETH_USDC</code>	ezETH collateral, borrowable: USDC
4	<code>LBTC_cbBTC</code>	LBTC collateral, borrowable: cbBTC
5	<code>wrsETH_WETH_wstETH</code>	wrsETH collateral, borrowable: WETH/wstETH
6	<code>wrsETH_USDC</code>	wrsETH collateral, borrowable: USDC
7	<code>weETH_WETH</code>	weETH collateral, borrowable: WETH
8	<code>wstETH_WETH</code>	wstETH collateral, borrowable: WETH
9	<code>cbETH_WETH</code>	cbETH collateral, borrowable: WETH
10	<code>cbBTC_USDC_GHO</code>	cbBTC collateral, borrowable: USDC/GHO
11	<code>syrupUSDC_USDC_GHO</code>	syrupUSDC collateral, borrowable: USDC/GHO

ID	Name (as encoded in address book)	Interpretation (not independently verified against live parameters)
12	<code>AAVE__USDC_GHO</code>	AAVE collateral, borrowable: USDC/GHO
13	<code>EURC__USDC_GHO</code>	EURC collateral, borrowable: USDC/GHO
14	<code>LBTC__USDC_GHO</code>	LBTC collateral, borrowable: USDC/GHO
15	<code>tBTC__USDC_GHO</code>	tBTC collateral, borrowable: USDC/GHO

The `collateral__borrowable` naming convention is consistent with the v3.7 isolated-eMode design described in Section 5.3 (a narrow, mostly single-collateral-asset category with specific permitted borrow assets), rather than the older multi-asset "ETH-correlated" or "stablecoins" category style described generically in Aave's own risk-parameters documentation (github.com/aave/risk-v3), which describes the historical two-category [stablecoins, ETH-correlated] model as of its own writing — that document is older/generic and does not reflect Base's actual 15-category configuration). **Whether each category is flagged "isolated" under v3.7, and the exact LTV/LT/LB values within each category, are live parameters — [LIVE-QUERY REQUIRED], see Section 12.**

7.5 Recently observed supply/borrow figures (illustrative snapshot, NOT a certified current state)

Retrieved via a cached secondary aggregator (Aavescan, aavescan.com/base-v3), snapshot dated **11 September 2026** per the page's own daily-snapshot table) and via a cached search snippet of the official Aave app markets page (app.aave.com/markets/?marketName=proto_base_v3), exact snapshot date uncertain but appears to be from the same general period based on the search index):

Asset	Supplied (USD, Aavescan 11 Sep 2026)	Borrowed (USD)	Supply APR	Borrow APR
cbBTC	\$211.9M	\$8.12M	0.01%	0.69%
WETH	\$206.6M	\$177.7M	1.75%	2.19% (incl. a disclosed Merkle incentive: 2.39% protocol + 0.20% reward split)
USDC	\$183.3M–\$183.4M	\$159.3M–\$163.6M (two snapshots differ slightly by hour)	3.55– 3.75%	4.54–4.66%
cbETH	\$127.6M	\$0.55M	~0%	0.07%
weETH	\$41.3M	\$7.1K	~0%	1.00%
wrsETH	\$38.8M	— (no active borrow reported)	~0%	—
wstETH	\$33.8M	\$1.05M	~0%	0.02%
syrupUSDC	\$10.8M–\$16.6M (two sources differ — see note)	— / \$0	~0%	—
EURC	\$8.11M–\$8.33M	\$4.05M	1.73– 1.87%	4.06–4.16%
LBTC	\$1.52M–\$1.53M	—	0%	—
GHO	\$1.41M–\$1.42M	\$1.26M–\$1.27M	3.54– 3.68%	4.44–4.58%
tBTC	\$0.45M–\$0.50M	\$0.057M	0.14– 0.16%	1.53–1.62%
AAVE	\$0.43M–\$0.46M	—	0%	—
USDbC	\$0.155M–\$0.157M	\$0.112–0.113M	2.86– 3.01%	7.97–8.46%

Asset	Supplied (USD, Aavescan 11 Sep 2026)	Borrowed (USD)	Supply APR	Borrow APR
ezETH	\$0.0205M	—	0%	—

Total market size (Aavescan, 11 Sep 2026): ~\$866.4M total supplied, ~\$356.5M total borrowed, i.e., **~41% aggregate utilization** at the market level (individual reserve utilization differs sharply — USDC alone shows ~87% utilization on its own dedicated page).

Interpretive cautions, required by the analytical rules in Section 1:

- The two-source figures disagree slightly (e.g., syrupUSDC \$10.8M vs. \$16.6M; USDC borrowed \$159.3M vs. \$163.6M) because they were captured at different hours/via different aggregation methods (Aavescan's own daily/hourly snapshots vs. a cached search-engine copy of the Aave app). This is exactly the kind of **regime-dependent, snapshot-sensitive figure** the Foundation Document warned against treating as a fixed fact — a production pipeline must timestamp every pull to the minute and reconcile against the Pool contract's own reserve data directly, not rely on either secondary aggregator as the source of truth.
- Several reserves show "0%" or "~0%" supply APR with essentially no active borrowing (AAVE, LBTC, ezETH, wrsETH) — this reflects low utilization on those specific reserves at the time of the snapshot, not that supplying carries zero risk; oracle, smart-contract, and freeze/pause risk remain regardless of realized yield.
- (cbBTC) is flagged as "(Disabled)" for borrowing in at least one cached view alongside weETH, cbETH, tBTC, and USDbC — this likely reflects a temporary borrow-freeze state (possibly a risk-management action, possibly a routine freeze) rather than a permanent protocol feature; **the specific reason and the exact set of currently-frozen/paused reserves is a live parameter, not captured with certainty by this snapshot.** [LIVE-QUERY REQUIRED]

7.6 What Section 7 explicitly does NOT establish

This section does **not** establish current LTV, liquidation threshold, liquidation bonus, supply cap, borrow cap, debt ceiling, or reserve factor for any Base asset. Despite extensive search, no fetchable, dated, Base-specific primary source returning these exact live values was found in this research pass (the official Aave app is a client-rendered SPA that returns no static values to a plain HTTP fetch; Basescan's contract-read UI likewise requires a live RPC call this environment cannot make; third-party scraper APIs exist — e.g., an Apify "Aave V3 Markets Scraper" advertising exactly these fields — but were not independently run or verified here).

Every one of these seven parameters, for every one of the 15 Base reserves, is marked [LIVE-QUERY REQUIRED]; see Section 12 for the exact contract and function to call.

8. Asset-by-asset oracle dependencies

8.1 General architecture confirmed for Base

Per Section 6.1–6.2, the path is: Base Aave reserve → `AaveOracle` (`0x2Cc0Fc26eD4563A5ce5e8bdcfe1A2878676Ae156`) → the per-asset "oracle-configured source" address listed in Section 7.2 → (for most non-stable, non-flat-BTC assets) either a raw Chainlink `AggregatorProxy`, or one of Aave's own `PriceCapAdapterBase`/`PriceCapAdapterStable`/composed-feed contracts, which themselves read one or more upstream Chainlink feeds.

8.2 What could NOT be determined per-asset in this pass

For **none** of the 15 Base reserves could this research independently confirm, from a fetchable primary source, whether the address-book "oracle-configured source" is (a) a raw Chainlink proxy, (b) an Aave `PriceCapAdapterBase` instance, (c) an Aave `PriceCapAdapterStable` instance, or (d) a composed feed — this requires reading the bytecode/verified source of each specific address on Basescan (or calling a `describe()`/similar view function if the adapter exposes one), which this research environment could not do (no RPC/Basescan-API access; Basescan's page could not be statically fetched with real contract-read data — see Section 1.3). **This is the single largest concrete gap in the oracle-dependency evidence chain and should be the first live-verification task before any oracle-based risk signal is computed.** [LIVE-QUERY REQUIRED for every row below]

8.3 Reasoned inference by asset class (inference, not verified fact — flagged accordingly)

Asset class	Base assets	Plausible adapter type (inference from Aave's documented adapter taxonomy, Section 6.2)	Confidence
Native ETH-pegged wrapper	WETH	Likely a raw Chainlink ETH/USD proxy (or an Aave/SVR-dedicated ETH/USD proxy per Section 6.3)	Low-to-moderate — plausible but not confirmed; the discrepant address found in Section 8.5 makes even this uncertain until resolved

Asset class	Base assets	Plausible adapter type (inference from Aave's documented adapter taxonomy, Section 6.2)	Confidence
Rebasing/exchange- rate LSTs	cbETH, wstETH, weETH, ezETH, wrsETH	Likely <code>PriceCapAdapterBase</code> (rate-capped), consistent with Aave's documented LST adapter family and its explicit inclusion of "weETH," "ezETH" as named examples in the adapter README	Moderate
BTC-pegged wrapped assets	cbBTC, tBTC, LBTC	Mixed / uncertain. The LBTC Ethereum precedent (Section 6.2) shows Aave migrating LBTC specifically to a ratio-aware CAPO feed because of its yield-bearing redesign — LBTC's Base oracle address is distinct from cbBTC/tBTC's, consistent with (but not proof of) a similarly distinct ratio-aware treatment on Base. cbBTC and tBTC share an oracle address on Base (Section 8.5) — if that shared address is a flat BTC/USD feed, tBTC would be priced as if perfectly redeemable 1:1 for BTC with no adapter accounting for any tBTC/BTC redemption-ratio divergence, which is a materially different (and potentially understated) risk treatment than what Aave has already done for LBTC on Ethereum for a structurally similar concern.	Low — this is a genuine open question requiring code-level verification, not a settled fact either way
Fiat-pegged stablecoins	USDC, USDbC, EURC, GHO	Likely <code>PriceCapAdapterStable</code> (fixed-cap), consistent with Aave's documented stablecoin	Moderate for USDC/USDbC/EURC; GHO is a distinct case

Asset class	Base assets	Plausible adapter type (inference from Aave's documented adapter taxonomy, Section 6.2)	Confidence
		adapter family explicitly naming USDC/EURC as examples	— as an Aave-native, over-collateralized, DAO- governed stablecoin (Section 11.3), its price- feed design and what it is actually capped against was not independently confirmed in this pass
Governance/utility token	AAVE	Likely a raw or Aave/SVR- dedicated AAVE/USD proxy	Moderate
Yield-bearing stable vault token	syrupUSDC (Maple Finance)	Likely a ratio/NAV-aware feed given it is a yield-accruing wrapped-stablecoin vault token, not a flat \$1 asset, but the specific mechanism was not confirmed	Low

8.4 Decimals

The address book explicitly encodes underlying-token decimals (not oracle-answer decimals, which are a separate, typically 8-decimal Chainlink convention and were not independently confirmed per feed in this pass):

WETH/cbETH/wstETH/weETH/ezETH/wrsETH/GHO/AAVE/tBTC = 18 decimals;

USDC/USDbC/EURC/syrupUSDC = 6 decimals; cbBTC/LBTC = 8 decimals. **Do not conflate underlying-token decimals with oracle-answer decimals when normalizing price data — this is a documented Aave/Chainlink integration pitfall the Foundation Document already flagged in principle (its Section D); this pass did not find a Base-specific instance of this error, only re-confirms the general hazard applies to all 15 Base assets.**

8.5 Consistency check: two conflicting oracle addresses found

While researching Section 8, a direct full-file fetch of `aave-dao/aave-address-book`'s

`AaveV3Base.sol` returned:

- `WETH_ORACLE = 0x9dA00D23465282005DB222a441a663eE7B9dfCc8`

- `cbETH_ORACLE = 0xB6E9fbba669763BD811B10A761110C449824b5BB`

A separate search-cached snippet of `bgd-labs/aave-address-book`'s `AaveV3Base.sol` and its companion `AaveV3Base.ts` (also captured 11 September 2026 in this research pass) returned, for the same two assets:

- `WETH_ORACLE = 0x71041dddad3595F9CEd3DcCFBe3D1F4b0a16Bb70`
- `cbETH_ORACLE = 0x8e11Ad4531826ff47BD8157a2c705F5422Da6A61`

Both snapshots agree on every other address checked (e.g., `POOL_EXPOSURE_STEWARD = 0x4f1F3E32f20847a4c9F002882eB607aD750b6115` in both). **This is an unresolved, materially important discrepancy** — the two oracle addresses are not textual variants of the same value; they are different contract addresses entirely. Two explanations are equally plausible and neither was confirmed:

1. `bgd-labs/aave-address-book` is a stale mirror/fork whose search-index cache reflects an older commit than the `aave-dao/aave-address-book` canonical repository's current `main` (the `aave-dao` org appears to be the actively-maintained successor per the repository's own README framing) — in which case the `bgd-labs` addresses may represent **the WETH/cbETH oracle configuration as it existed at some earlier point**, and Aave governance changed the WETH/cbETH price source on Base at some later date (oracle migrations for major assets are a documented, routine governance action — see Section 9).
2. The search cache for one or both repositories is itself stale relative to the live repository content, independent of which org is canonical.

This must be resolved by a live call to `AaveOracle.getSourceOfAsset(WETH_UNDERLYING)` and `getSourceOfAsset(cbETH_UNDERLYING)` on Base before any WETH/cbETH oracle-dependency signal is computed. Until resolved, this pack treats the WETH and cbETH oracle source addresses as **UNKNOWN / NOT VERIFIED** despite having two specific candidate values, precisely because false precision (picking one of two plausible values without a tiebreaker) is exactly the failure mode Section 1.2 prohibits. No other asset's oracle address showed a similar discrepancy across the sources checked in this pass, but not every asset was cross-checked against a second independent source, so absence of a flagged conflict elsewhere is not proof of consistency.

8.6 Shared-source concentration findings (high confidence — directly from address-book data, not inference)

- **USDC and USDbC share the identical oracle-source address** (`0xf52D010c7d4ecBfda92c2509900593CE34535D86`). A single upstream failure, stale-price condition, or governance misconfiguration affecting that one adapter/feed would affect both reserves' pricing simultaneously and identically — this is a direct, verifiable oracle-concentration fact, not merely a plausible-sounding correlation, and is a stronger form of the "oracle

dependency/concentration" candidate signal the Foundation Document already flagged conceptually.

- **cbBTC and tBTC share the identical oracle-source address**

(`0x3A932b286715abc4A86a4ACAF68A6cdD89E0d446`). As discussed in Section 8.3, this is potentially more consequential than the USDC/USDbC case, because cbBTC (a Coinbase-custodied, closely-pegged wrapped BTC product) and tBTC (a decentralized, threshold-signature-bridged wrapped BTC product with a materially different trust model and historical redemption mechanics) are not obviously interchangeable from a redemption-risk standpoint, yet Aave's Base deployment appears to price them identically. **Whether this shared address is itself a ratio-aware composite feed that happens to output the same number for both (benign) or a flat BTC/USD feed applied uniformly regardless of each token's actual peg mechanics (a potential mispricing-of-divergence risk) is exactly the open question flagged in 8.3 and 8.2, and is the single highest-value oracle-architecture item for the next research/verification pass on this protocol.**

9. Governance and historical parameter changes

9.1 Version-upgrade history relevant to Base (dated, primary-sourced)

Date	Event	Source type
9 January 2026	Aave v3.6 changelog entry — "Liquid eModes" (assets listable in multiple eModes), stable-rate-logic removal; initial deployment list given as Sonic, Optimism, Gnosis, Scroll, ZKSync, Celo, Metis, Soneium, Ethereum EtherFi (Base not listed as part of this initial wave in the retrieved changelog text)	Primary (aave.com/docs/resources/changelog)
~16 February 2026 (ARFC posted)	"[ARFC] BGD. Aave v3.7" — feature proposal: eMode isolated flag, removal of legacy isolated collaterals/borrowable-in- isolation, removal of L2	Primary (governance.aave.com)

Date	Event	Source type
	sequencer oracle gating, append-only reserves list, liquidation-calculation refinements	
26–29 March 2026	v3.7 audit reports published: MixBytes (26 Mar), Sherlock (26 Mar), Pashov (27 Mar), Certora (29 Mar)	Primary (github.com/aave-dao/aave-v3- origin/tree/main/audits; aave.com/security)
Execution date not independently pinned in this pass ("Part 1")	"Upgrade Aave instances to v3.7 Part 1" executed on Sonic, Optimism, Gnosis, Scroll, Celo, MegaEth, XLayer, Ethereum (EtherFi) — Base explicitly excluded from Part 1	Primary (on-chain governance execution record, retrieved via cache)
~29 May 2026 (secondary- sourced date; not independently confirmed against the specific Base execution transaction)	v3.7 rollout stated complete across Ethereum (Core, Lido), Polygon, Avalanche, Arbitrum, Base , BNB Chain, Linea, Plasma, Mantle — this is presumed to be "Part 2"	Primary changelog states completion; exact Base execution date is secondary-sourced only
Ongoing as of this review (dated within the review window)	"[ARFC] Unified Handling of the Isolated Flag in Aave V3.7 E-Modes" — active discussion proposing which eMode categories DAO-wide (not confirmed to include or exclude any specific Base category) should be flagged isolated	Primary (governance.aave.com), status: open discussion, not yet an executed decision as retrieved

Date	Event	Source type
1 week before this review (per search-result age metadata)	Secondary trade-press report: "Aave Governance Approves Base Parameter Update For v3 Markets" — described only in general terms (eMode optimizations and collateral caps); the specific parameters changed were not identified by this research	Secondary (bitcoinist.com) — treat as a signal that a Base-specific parameter-change proposal passed very recently, but the content is UNKNOWN / NOT VERIFIED from this source alone; the underlying AIP/forum thread should be located and read directly

9.2 Original Base deployment proposal

The original Base listing process began with "[ARFC] Aave V3 Deployment on Base," authored by Flipside Crypto ([@fig](#)), dated 15 June 2023, proposing deployment upon Base mainnet launch. This predates and is superseded by all of the version-upgrade history above; it is recorded here only for completeness of the deployment's governance lineage and was not read in full detail in this pass (only the forum-thread summary snippet was retrieved).

9.3 Ongoing/automated parameter-change infrastructure

See Section 5.5 for the Risk Steward / Edge Risk Oracle / [AaveStewardsInjector](#) automation layer, confirmed deployed on Base with its own distinct contract addresses. **Historical practice on other Aave deployments** (per [aave-dao/aave-proposals-reports](#), a primary GitHub repository indexing security reports for on-chain governance proposals) shows this kind of automated/constrained cap, rate, and collateral-parameter update is extremely frequent across Aave's full multi-chain footprint — the repository's own index lists dozens of "Caps update," "Risk params update," and asset-specific parameter-change entries across Ethereum, Polygon, Arbitrum, and Optimism markets, illustrating the general cadence of this kind of change even though a Base-specific enumeration was not extracted line-by-line in this pass. **A risk pipeline should treat "no forum thread found for a given date" as weak evidence of no change, not as proof, given this automated layer exists.**

9.4 What remains unresolved

- No complete, dated list of every Base-specific asset listing, cap change, freeze, or oracle change was compiled in this pass; Section 9.1's table is representative of what this pass found, not exhaustive.
- The exact isolated-flag configuration for each of Base's 15 eMode categories (Section 7.4) was not determined.

- The specific content of the "Base Parameter Update" reported by secondary press roughly one week before this review was not independently verified.

10. Audits, incidents and failure evidence

10.1 The Kelp DAO / rsETH bridge exploit (18 April 2026) — protocol-wide incident with confirmed Base risk-containment actions

This is the most significant Aave-adjacent security event identified in this research pass and is **not Base-specific in origin**, but **did produce confirmed Base-specific emergency governance actions**, so it must be included and carefully scoped.

What happened (facts, multiply-corroborated across primary Aave governance posts and reputable secondary reporting — CoinDesk, The Defiant, Forbes, CryptoBriefing, KuCoin, CryptoTimes, NYDIG research, CryptoSlate, Phemex, dated 18–30 April 2026):

- At approximately 17:35 UTC on 18 April 2026, an attacker exploited a cross-chain message-verification flaw in Kelp DAO's LayerZero-powered bridge for its rsETH liquid-restaking token, minting/releasing **116,500 rsETH (~\$292–293 million, ~18% of rsETH's circulating supply)** without corresponding backing.
- The attacker deposited a portion of this unbacked rsETH (89,567 rsETH per Aave's own incident report) as collateral on **Aave V3 and Aave V4**, on **Ethereum and Arbitrum**, and borrowed real assets against it — approximately 52,834 WETH on Ethereum and 29,782 WETH + 821 wstETH on Arbitrum (figures per Chaos Labs' founder as quoted in Forbes; Aave's own incident report frames the total as ~82,650 WETH + 821 wstETH borrowed against the attacker's positions across Ethereum Core and Arbitrum).
- **Aave's own public position, stated by founder Stani Kulechov and the official Aave account, is that this was not a vulnerability in Aave's own contracts** — the exploit originated in KelpDAO/LayerZero infrastructure external to Aave; Aave's systems (freezing, LTV-to-zero) are described as having "functioned as designed" once the bad collateral was recognized as a threat.
- Aave's own incident report (via LlamaRisk, dated 20 April 2026, cited in multiple secondary sources) modeled potential bad debt at **\$123.7 million to \$230.1 million**, depending on how Kelp DAO ultimately allocates the shortfall across chains (a mainnet-absorbs-all-losses scenario vs. an L2-concentrated-losses scenario, the latter falling most heavily on Arbitrum and Mantle).
- **Confirmed Base-specific actions:** per CryptoBriefing (dated 21 April 2026, itself synthesizing Aave's own incident timeline), Aave froze all rsETH and wrsETH reserves across its V3 deployments on 18 April 2026 and set their LTV to zero; on 19–20 April 2026 it adjusted

WETH interest-rate models across multiple chains and then **froze WETH on Core, Prime, Arbitrum, Base, Mantle, and Linea** specifically to stop fresh borrows and limit spillover into other reserves. **Base holds a listed wrsETH reserve (Section 7.2) and this WETH freeze is a confirmed, Base-specific defensive governance action taken in direct response to a protocol-wide incident that did not originate on Base and did not (per available evidence) involve any Base-specific attacker collateral position.**

- A root-cause governance post-mortem (TEMP CHECK, "Post-rsETH Collateral Framework: Tier-Based LTV Reductions and Wrap-Depth Ineligibility Limits," governance.aave.com, dated 24 April 2026, primary) attributes part of the exposure to a January 2026 governance decision (Proposal 311) that had activated a 93% LTV / 95% liquidation-threshold eMode for rsETH three months before the exploit, explicitly to compete for rsETH TVL against other lending markets, without (per the same post) a bridge-risk assessment being conducted at listing time.
- A large, multi-protocol, multi-DAO recovery effort ("DeFi United") was organized starting 23 April 2026, with Aave DAO service providers proposing a 25,000 ETH contribution, founder Stani Kulechov personally pledging 5,000 ETH, and additional commitments from Mantle Network (a 30,000 ETH credit facility via its MIP-34 proposal), ether.fi, Consensys/Joseph Lubin, and others; total pledged commitments were reported around \$161–314 million (figures vary by reporting date as the total grew). On-chain analyst figures cited in secondary press put the total shortfall at **75,081 ETH**, later reduced toward roughly a 5,505 ETH residual gap after recoveries, freezes (including 30,766 ETH frozen by the Arbitrum Security Council and 43,168 ETH recovered from Kelp), and DAO contributions.
- **This pack could not confirm a final, closed-out settlement figure or date as of 11 September 2026** — the most recent evidence retrieved dates to late April/early May 2026 (Mantle governance vote in progress). Whether the DAO votes, the Umbrella-vault slashing question (Section 5.4), and the final bad-debt write-down were fully resolved by the time of this review is **UNKNOWN / NOT VERIFIED** and should be checked directly against the Aave governance forum before use.

Distinguishing Base-specific vs. protocol-wide, as required: the exploit's origin (Kelp DAO/LayerZero), the attacker's collateral deposits, and essentially all of the reported bad-debt exposure were on **Ethereum and Arbitrum, not Base**. Base's only confirmed direct involvement is (a) holding a wrsETH reserve that was frozen as a precaution, and (b) having its WETH reserve frozen as a precaution against spillover. No evidence in this pass indicates Base itself carried attacker-originated bad debt from this incident.

10.2 Audit history for the deployed protocol version (v3.7)

Auditor	Date	Scope
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MixBytes	26 March 2026	Aave v3.7
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Auditor	Date	Scope
Sherlock	26 March 2026	Aave v3.7
Pashov	27 March 2026	Aave v3.7
Certora	29 March 2026	Aave v3.7 (formal verification)

Earlier-version audit history (for context on cumulative review depth, not all applicable to the currently-deployed v3.7 code): v3.6 — Blackthorn (16 Nov 2025), Certora (18 Nov 2025), MixBytes (18 Nov 2025), Savant (18 Nov 2025), Pashov (29 Nov 2025); v3.5 — ABDK, StErMi, MixBytes (all mid-July 2025), Certora (14 Jul 2025); v3.3 — Certora, StErMi, Oxorio, Sherlock (Jan 2025); v3.2 — Certora, Enigma Dark, Oxorio, Pashov (Sep 2024); v3.1 — Certora, MixBytes, Cantina competition, internal SterMi review (Apr 2024); v3.0.2 — SigmaPrime, Certora (Apr 2023); v3.0.1 — PeckShield, SigmaPrime (Dec 2022); original v3 — ABDK, OpenZeppelin, Trail of Bits, PeckShield, SigmaPrime, Certora. All from [\[aave-dao/aave-v3-origin/audits\]](https://github.com/aave-dao/aave-v3-origin/audits) (primary GitHub directory) and [\[aave.com/security\]](https://aave.com/security) (primary), both retrieved in this pass. **A published audit is evidence of review, not a certification against unknown-unknowns or against bugs introduced by later configuration/governance actions (e.g., the rsETH incident above involved no code vulnerability at all) — see Section 15 for this exact caution already required by the analytical rules.**

10.3 Bug bounty

Aave's own security page states a community bug bounty program exists; one secondary source (a general audit-checklist article, not Aave-published) cites a "\$250K rewards" figure — **this specific number is from a secondary, non-Aave source and was not independently confirmed against Aave's own bug-bounty program terms in this pass.**

10.4 Base-chain-level (not Aave-specific) reliability incidents, 2024-2026

These are chain-level dependencies relevant to any Base-deployed protocol's liveness and to the Chainlink L2-sequencer-feed discussion (Section 6.1), not Aave-specific incidents:

Date	Incident	Duration	Nature	Source
September 2024	Sequencer-related halt	~17 minutes	Block-production stall	Secondary (CoinPaprika, referencing this as a prior data point)
			Disputed characterization: Base's own status page reported halted block production, but per an independent analysis (Metrika), blocks were in fact still being produced during this window, just nearly empty of user transactions — i.e., a throughput/activity collapse that was mis-signalized as full downtime by a coarse status indicator, not a true production halt	
5 August 2025	Sequencer-related stall	~30–33 minutes		Secondary (Metrika post-mortem analysis; CoinPaprika)
29–31 May 2026	TEE (Trusted Execution Environment) enclave failure halting the state-update/proposal system that posts Base's commitments to Ethereum	~36 hours (Base's own status page) / "~30 hours" (secondary paraphrase)	Settlement/withdrawal delay, not a block-production halt — ordinary transactions and dApp interactions continued normally throughout; only L1 state commitments and the withdrawal-proof pathway were affected; masked from most users because Base's standard withdrawal challenge period is already 7 days. Linked to a recent "Azul" upgrade that reportedly	Secondary, but citing a primary source: "Base official status page — Mainnet Withdrawal Delay, 29–31 May 2026" (page itself not independently fetched in this pass)

Date	Incident	Duration	Nature	Source
			moved Base toward an independent proving architecture separate from the shared Optimism Superchain stack — this architectural claim is from secondary reporting only and was not verified against Base's own technical documentation in this pass. No funds lost; first production failure of the new proving architecture per the same reporting.	
25-26 June 2026	Two back-to-back sequencer block-production halts, same root cause: a bug in sequencer block-building logic left stale execution-journal state (accessed accounts/storage slots) uncleared after a failed transaction validation, causing the next block to be built on inconsistent state	First incident: ~108-116 minutes (11:47am EDT 25 June to resolution); second: ~13-20 minutes (11:28am EDT 26 June)	True block-production halts — no new blocks at all during the windows; chain integrity was not compromised and all funds remained safe per Base's own statement; recovery required manual patching and, for the second incident, resolving a race condition in sequencer catch-up	Primary: Base's own engineering post-mortem, blog.base.dev/postmortem-june-25th-block-production-outage, corroborated by extensive secondary reporting (CoinPaprika, Cryptopolitan, DEXTools News, SpotedCrypto, Grafa, CryptoDaily, Field Elevate)

Aggregate pattern, stated plainly as an interpretation (not a certified fact): Base has experienced at least four distinct chain-level reliability incidents across roughly 22 months (Sept 2024 – June 2026), of at least three qualitatively different failure modes (short sequencer stall, mis-signaled activity collapse, settlement/withdrawal-proof delay, and a genuine multi-incident block-production halt), and runs, as of the most recent (June 2026) incident reporting, a single centralized sequencer operated by Coinbase with a Stage-1 decentralization rating per L2Beat (a data point cited by secondary sources, not independently re-verified against L2Beat directly in this pass) and no automatic sequencer failover.

10.5 Why Section 10.4 interacts directly with Section 5.2 (the v3.7 sequencer-sentinel removal)

Aave V3 Base's confirmed v3.7 upgrade **removed the protocol-level sequencer-uptime gating (`PriceOracleSentinel`) that, in earlier Aave versions, would delay new liquidations for a grace period after a detected L2 sequencer outage.** Base experienced two genuine, multi-incident block-production halts in June 2026 (Section 10.4) — after the v3.7 rollout is stated to have completed (29 May 2026, per the secondary-sourced date in Section 9.1). **This research could not determine, and flags as an important open question for the next verification pass, whether Aave V3 Base's liquidation logic behaved any differently than normal immediately before/after the June 2026 outages** (e.g., whether stale prices persisted through the halt and could have produced liquidations against prices that had not updated during the outage window, now that the sentinel-based grace-period protection has been removed at the protocol level). No evidence of any actual mispriced liquidation, bad debt, or user harm on Base tied to the June 2026 outages was found in this research — this is flagged as a **structural risk condition worth monitoring**, not a confirmed incident.

10.6 Other historical Aave incidents referenced but not independently re-verified in this pass

The Foundation Document's own review did not catalogue Aave's pre-2026 incident history (e.g., historical UST-related freezes referenced only incidentally in an old newsletter search result, or general "stETH depeg" parameter responses from 2022 found in this pass's search results as background). These are **not reproduced as findings here** because they predate the review's practical relevance window and were only encountered as incidental search artifacts, not deliberately researched; a dedicated historical-incident pass (Aave Safety Module activations, any pre-2026 bad-debt events, the 2022 UST freeze, etc.) remains a genuine gap — see Section 16.

11. RWA/stablecoin reserve-integrity evidence

11.1 GENIUS Act implementation status

See Section 4.2 in full. Summary for this section's purpose: **as of 11 September 2026, no GENIUS Act final rule exists and no issuer has been approved under it.** Any RWA/stablecoin scoring input that references "GENIUS Act compliance" or "GENIUS Act approval" as a current, checkable legal status is referencing something that does not yet exist and must be marked accordingly.

11.2 Circle-issued stablecoins on Base's Aave reserves (USDC, EURC, and legacy USDbC)

The Foundation Document already identified circle.com/transparency as "a starting index for USDC reports, not an instrument-level conclusion" (its footnote [^F7]-adjacent guidance in its "Important missing material" section). This pass did not re-fetch Circle's live attestation reports or reserve-composition disclosures in detail; the following is carried forward as unresolved:

- **UNKNOWN / NOT VERIFIED in this pass:** current USDC and EURC reserve composition, current attesting accounting firm, most recent attestation date, current issuer legal-entity structure (Circle Internal Financial Statements vs. Circle's regulated-entity trust/money-transmitter licenses by state), and whether USDbC (the legacy Base-bridged USDC representation, distinct from native USDC) carries any different or additional risk characteristics beyond being a bridged, not natively-issued, representation. USDbC's declining reserve size on Aave Base (Section 7.5: ~\$0.155M supplied) suggests market-driven migration away from the bridged representation toward native USDC, consistent with Circle's own native-USDC-on-Base rollout, but this migration dynamic itself was not independently documented with dates in this pass.
- **Legal-basis point (new, from Section 4.2):** Circle's USDC/EURC currently operate under pre-GENIUS-Act state trust-company/money-transmission licensing and Circle's own attestation practices, not under any GENIUS Act PPSI license (none exist yet). A risk methodology should track Circle's **actual current regulatory status** (state licenses, any NYDFS or equivalent supervision, attestation cadence and auditor) as the presently operative legal/disclosure regime, and treat "GENIUS Act eligibility once rules finalize" as a separate, forward-looking, non-current data point.

11.3 GHO — Aave's native stablecoin (materially different risk category from USDC/EURC)

New research this pass, primary-sourced (aave.com/docs/ecosystem/gho,

stablecoin/facilitators):

- GHO is **not** a fiat-reserve-backed stablecoin. It is minted through a "facilitator" model: any Aave-governance-approved contract or entity (a "Facilitator") can mint/burn GHO up to a DAO-assigned "bucket capacity"; the **primary facilitator is the Aave V3 Ethereum Market itself** — ordinary users mint GHO by borrowing it against over-collateralized crypto/RWA collateral inside Aave, the same way they would borrow USDC or WETH.
- A second facilitator type, the **GHO Stability Module (GSM)**, lets users swap other stablecoins (USDC, USDT, in some descriptions) for GHO near 1:1 minus a small fee, acting as a peg-defense mechanism.
- GHO's peg stability mechanism is **primarily the governance-set borrow interest rate** (raised/lowered by the DAO to manage supply-demand pressure on the peg), not a reserve-asset redemption guarantee. There is **no centralized issuer, admin key, or off-chain reserve custodian for GHO** — this is structurally opposite to USDC/EURC's model, and a risk methodology must not apply the same reserve-attestation-based evaluation framework to GHO that it would apply to Circle's stablecoins; GHO's risk surface is instead the collateral quality and liquidation mechanics of whatever backs each individual GHO mint (i.e., GHO risk is largely a restatement of Aave protocol risk itself, not an independent instrument risk).
- **This pass could not confirm whether Base has its own local GHO facilitator/minting capability, or whether Base's GHO reserve (Section 7.2) functions purely as a borrow/supply market for GHO that was minted elsewhere (e.g., on Ethereum) and bridged to Base.** Base's GHO oracle-source address is distinct from any other asset's (Section 7.2), which is at minimum consistent with GHO needing its own price-cap treatment against \$1, but the underlying facilitator/bridging architecture for GHO specifically on Base is **UNKNOWN / NOT VERIFIED** in this pass. General Aave documentation elsewhere (not independently confirmed for Base specifically) references a "RemoteGSM" cross-chain GHO architecture using CCIP bridging for other chains (e.g., a Monad GSM referenced in an app.aave.com governance snippet retrieved in this research) — whether Base uses an equivalent mechanism was not confirmed.

11.4 What issuer-specific documentation would be required before scoring an actual RWA or stablecoin (methodology checklist, not yet fulfilled for any Base asset)

Per the Foundation Document's "RWA inference" boundary (its Section B) and this pass's regulatory findings, a defensible instrument-level RWA/stablecoin risk assessment would require, at minimum, none of which were obtained for any specific Base-listed asset in this pass:

1. **Legal entity and governing law.** The exact issuing legal entity (not just a brand name), its jurisdiction of organization, and which regulator(s) currently supervise it (state money-

transmitter licenses, a trust-company charter, a bank charter, or — once finalized — GENIUS Act PPSI/FQPSI/SQPSI status).

2. **Binding redemption terms.** The actual, current terms and conditions or trust agreement governing a holder's (or, critically, a wrapped/bridged/Aave-supplied holder's) legal right to redeem the token for the underlying reference asset — including whether an Aave supplier of, e.g., USDC actually has any direct redemption right at all, or only a claim against the Aave pool (this is a distinct legal question the Foundation Document already flagged conceptually and which remains unresolved for every Base asset specifically).
3. **Custody and segregation evidence.** Where reserve assets are actually held, under what custodial/trust arrangement, and whether they are legally segregated from the issuer's general estate (relevant under both current state trust law and the GENIUS Act's insolvency provisions once effective).
4. **Most recent attestation or audit report,** the identity and independence of the attesting/auditing firm, its scope (attestation vs. full audit — a materially different assurance level), and its date relative to the review date.
5. **Reserve composition detail,** matched to the GENIUS Act's eligible-asset categories (Section 4 of the statute, carried forward in Section 4.1 above) even where the issuer is not yet legally bound by them, as a useful comparative benchmark.
6. **Disclosed fees, notice periods, and any regulator-imposed discretionary redemption limitations.**
7. **Technological capability and history of complying with lawful orders** (freeze/seize/blacklist function existence, and any historical instances of its use), relevant both to the GENIUS Act's AML/sanctions provisions and to a general "can this token be involuntarily frozen" risk signal independent of GENIUS Act status.
8. **For non-fiat-backed instruments (GHO and any future Aave-native or RWA-collateralized facilitator issuance):** the specific collateral basket, facilitator bucket capacities and current utilization, and the peg-defense mechanism's historical performance (e.g., has GHO's peg ever traded materially away from \$1, and what governance action, if any, restored it) — none of which were compiled for GHO specifically in this pass beyond the mechanism description in 11.3.

None of items 1–8 were completed for USDC, EURC, USDbC, or GHO as Base-specific, instrument-level dossiers in this pass; this is recorded as an explicit gap in Section 16, not silently assumed satisfied.

12. Available live/on-chain data

This section gives the exact retrieval method for every parameter this pack could not verify statically, so the next research or engineering pass can close the gap directly rather than re-

deriving the address list from scratch.

Data needed	Contract	Function (Aave <code>IPoolDataProvider</code> <code>IPoolConfigurator</code> <code>AAVE</code>)
LTV, liquidation threshold, liquidation bonus, decimals, reserve factor, usage-as-collateral flag, borrowing-enabled flag, stable-borrow flag, active/frozen/paused flags	<code>AAVE_PROTOCOL_DATA_PROVIDER</code> (<code>0x0F43731EB8d45A581f4a36DD74F5f358bc90C73A</code>)	<code>getReserveConfigurationData(assetId)</code>
Supply cap, borrow cap	same contract	<code>getReserveCaps(assetId)</code>
Debt ceiling (isolation mode)	same contract	<code>getDebtCeiling(assetId)</code>
eMode category membership, isolation flag, per-category LTV/LT/LB	<code>POOL</code> (<code>0xA238Dd80C259a72e81d7e4664a9801593F98d1c5</code>)	<code>getEModeCategoryData(uint8 id)</code> Section 7.4 (exact v3.7 struct flag field, should be confirmed by <code>Pool</code> implementation's ABI, since changed between v3.6 and v3.7)

Data needed	Contract	Function (Aave <code>IPoolDataProvider</code> / <code>IPoolConfigurator</code> / <code>AAVE</code>)
Which price source each asset actually uses	<code>ORACLE</code> / <code>AaveOracle</code> (<code>0x2Cc0Fc26eD4563A5ce5e8bdcfe1A2878676Ae156</code>)	<code>getSourceOfAsset(address asset)</code>
Live price	same contract	<code>getAssetPrice(address asset)</code> <code>assets)</code> (batch)
Underlying feed decimals, heartbeat/deviation (if a raw Chainlink proxy)	The specific address returned by <code>getSourceOfAsset</code>	<code>decimals()</code> , <code>description()</code> , <code>heartbeat/deviation</code> are not used by most Chainlink OCR aggregators against Chainlink's own public specific proxy address once
Reserve balances, total supply/debt, liquidity index, variable borrow index	<code>POOL</code>	<code>getReserveData(address asset)</code>
Current interest-rate-strategy parameters (optimal utilization, slopes)	The per-asset <code>INTEREST_RATE_STRATEGY</code> address (Section 7.2)	Strategy-contract-specific variables: <code>getOptimalUsageRatio()</code> , <code>getVariableRateSlope2()</code> , <code>getDefaultInterestRateStrategyV</code>
Active governance roles / who can act	<code>ACL_MANAGER</code> (<code>0x43955b0899Ab7232E3a454cf84AedD22Ad46FD33</code>)	<code>isPoolAdmin</code> , <code>isRiskAdmin</code> , <code>is</code> address

Data needed	Contract	Function (Aave <code>IPoolDataProvider</code> / <code>IPoolConfigurator</code> ABI)
Raw parameter-change event history	<code>POOL_CONFIGURATOR</code> (<code>0x5731a04B1E775f0fdd454Bf70f3335886e9A96be</code>)	Event logs: <code>CollateralConfigured</code> , <code>BorrowCapChanged</code> , <code>SupplyCapChanged</code> , <code>ReserveFrozen</code> , <code>ReservePaused</code> etc. (standard <code>IPoolConfigurator</code> events)
Liquidation event history	same contract's underlying <code>Pool</code>	<code>LiquidationCall</code> event (signature <code>0xe413a321e8681d831f4dbccbca7</code> per a QuickNode sample-app on Ethereum mainnet — the event was added in Aave V3 deployments since its new interface)
Deficit/bad-debt recognition events	<code>Pool</code>	<code>DeficitCreated</code> and related variants (exact event name should be verified in deployed <code>Pool</code> ABI)
L2 sequencer status (informational only — confirmed not consulted by Aave's own liquidation logic post-v3.7, per Section 5.2/10.5)	Chainlink's Base Sequencer Uptime Feed (address not re-confirmed in this pass; carried forward from the Foundation Document's footnote [^F28], which should be re-verified live)	<code>latestRoundData()</code>
Base chain identity/liveness	Base RPC	<code>eth_chainId</code> (expect <code>0x2105</code> for Base)

13. Available historical/off-chain data

Source	What it provides	Base coverage confirmed?	Primary/secondary
Aave Pool / PoolConfigurator event logs (via any full-node/indexer, e.g., a self-hosted archive node or a paid RPC provider with log access)	Complete, ground-truth history of every reserve parameter change, liquidation, deficit event, freeze/pause, and eMode change since Base deployment	Yes — same contracts as Section 12	Primary
The Graph — Aave V2/V3/V4 subgraphs	Indexed lending-market data: positions, health factors, liquidations, flash loans, reserve history, governance	Yes — a community-maintained MCP server (PaulieB14/graph-aave-mcp , inspected in this pass) explicitly lists Base among 7 covered chains across 11+ subgraphs, alongside Ethereum, Arbitrum, Polygon, Optimism, Avalanche, Fantom	Secondary (community-indexed, not Aave published, though indexes Aave's own primary event data)

Source	What it provides	Base coverage confirmed?	Primary/secondary
Dune Analytics	Community SQL-queryable decoded Aave tables and published dashboards, including dedicated liquidation-analysis queries/dashboards (e.g., "Aave V3 Liquidation Aggregated," "Aave Liquidations Overview")	Not confirmed Base-specific for the exact dashboards spot-checked in this pass (several appeared Ethereum-focused), but Dune's underlying decoded-table infrastructure generally covers any chain it indexes, which includes Base	Secondary (community-built on top of primary on-chain data)
Aavescan (aavescan.com)	Live and historical (daily-snapshot, with hourly/full-history behind a paid "Pro" tier) supply, borrow, APR, and utilization figures per asset per market, including a dedicated Base V3 market page and per-asset sub-pages	Yes — directly confirmed in this pass (Section 7.5)	Secondary (aggregator) — cited by academic (NYU Stern, Ethereum Foundation-adjacent researchers per its own "About" page and financial-preferences sources (Bloomberg) as a data reference
Official Aave app markets page (app.aave.com/markets/?marketName=proto_base_v3)	The canonical first-party live reserve list and figures	Yes	Primary

Source	What it provides	Base coverage confirmed?	Primary/secondary
Aave's own documented Markets API/SDK	First-party structured market/reserve data schema (the <code>Market</code> , <code>Reserve</code> , <code>EmodeMarketCategory</code> TypeScript interfaces)	Documented generally, not confirmed against a live Base response in this pass	Primary (documentation); live responses not independently verified
<code>aave-dao/aave-proposals-reports</code> (GitHub)	Indexed security/diff reports for essentially every on-chain Aave governance proposal across all deployments, giving a structured historical register of listings, cap changes, freezes, rate changes, etc.	Confirmed to include multichain entries; Base-specific entries not individually enumerated in this pass but the repository's structure supports filtering by network	Primary

Source	What it provides	Base coverage confirmed?	Primary/secondary
Third-party commercial scraper APIs (e.g., an Apify "Aave V3 Markets Scraper" actor advertising LTV/liquidation-threshold/reserve-factor fields "direct from the public Aave data API" across 11 markets including Base)	Potentially exactly the current-parameter data this pack could not obtain directly	Advertised, not independently run or verified in this pass	Secondary, unverified
Chainlink historical round data (docs.chain.link/data-feeds/historical-data)	Proxy phase/round history for any Chainlink feed once its address is confirmed (Section 8.2's open item)	Mechanism confirmed generally (carried forward from the Foundation Document's footnote [^F29]); not exercised against a specific confirmed Base feed address in this pass because no Base feed address was code-verified in this pass	Primary
Basescan (basescan.org)	Contract source verification, read-contract UI, transaction/event history, token-holder analytics	Referenced extensively as the address book's own linking convention (every address in Sections 7–8 links to a basescan.org/address/... page in the source repositories) but not independently fetched with live contract-read data in this research environment	Primary (block-explorer indexing primary chain data)

Source	What it provides	Base coverage confirmed?	Primary/secondary
DeFiLlama	Chain-level and protocol-level TVL, a general secondary reference cited incidentally in this pass (e.g., Base chain TVL of ~\$4.04B cited by a secondary sequencer-outage article dated 26 June 2026; GHO circulating market cap figures cited by secondary GHO explainers)	General chain/protocol-level coverage confirmed by citation, not independently queried for a Base-Aave-specific breakdown in this pass	Secondary
Reuters/CoinDesk/The Defiant/Forbes/CryptoBriefing and similar reputable crypto-native and financial press	Incident timelines, dollar-figure estimates, governance-vote outcomes reported faster than forum threads are sometimes indexed	Confirmed useful in Section 10.1	Secondary

14. Candidate measurable risk signals

These extend the Foundation Document's own candidate-signal table (its Section E) with Base/Aave-V3.7/GENIUS-Act-specific additions found in this pass. As with the Foundation Document, these are candidate observations only — no weighting, threshold, or selection is proposed.

Candidate observation	Primary evidence required	Interpretation limits
Oracle-source concentration between nominally distinct assets	<code>AaveOracle.getSourceOfAsset()</code> for every reserve, compared pairwise (Section 8.6's USDC/USDbC and cbBTC/tBTC findings as concrete existence proof this signal is non-trivial on Base specifically)	A shared address does not by itself prove mispricing; it proves a shared single point of failure and, for economically dissimilar assets sharing a source (e.g., two BTC wrappers with different redemption mechanisms), raises a specific question about whether asset-specific risk (e.g., a bridge/custody failure unique to one wrapper) would be reflected in its price at all
Interest-rate-strategy contract sharing	<code>INTEREST_RATE_STRATEGY</code> address per reserve (Section 7.2's finding that 12+ of 15 Base reserves share one contract address)	Shared contract code is not shared parameters; must be paired with a live read of each reserve's actual configured curve inputs before drawing any conclusion about correlated rate behavior
Automated (Risk Steward/Edge Risk Oracle) parameter-change frequency and magnitude, vs. full-governance-vote parameter changes	<code>PoolConfigurator</code> event logs (Section 12), cross-referenced against <code>aave-proposals-reports</code> and governance-forum threads	A high frequency of small, bounded automated changes is a different risk profile from infrequent, large, fully-deliberated governance changes; conflating the two into one "parameter volatility" metric would obscure which mechanism produced a given change and its actual review depth
eMode isolated-flag status and its interaction with an	<code>Pool.getEModeCategoryData()</code> per category (Section 12), cross-referenced against the open ARFC in Section 5.3	A category not yet flagged "isolated" under the still-being-debated policy may

Candidate observation	Primary evidence required	Interpretation limits
asset's non-eMode collateral usage		carry a materially different collateral-contribution risk than one that is; this is a genuinely time-varying, currently-under-DAO-debate parameter, not a settled protocol constant
Chain-level (Base) liveness-incident frequency and type, and its temporal relationship to Aave's own v3.7 sequencer-sentinel removal	Section 10.4/10.5's compiled incident timeline, plus Aave's own liquidation-event logs during and immediately after each incident window	A liveness incident does not itself prove a mispriced liquidation occurred; correlating an actual <code>LiquidationCall</code> timestamp against a confirmed chain-outage window is required before treating "outage occurred" as "liquidations were affected"
Kelp DAO/rsETH-incident-style cross-protocol collateral contagion exposure (i.e., does a given Base reserve accept, as collateral, any token whose backing depends on a bridge or a third-party protocol's own solvency)	Per-asset dependency mapping: is the underlying token (wstETH, weETH, cbETH, ezETH, wrsETH, LBTC, tBTC, cbBTC, syrupUSDC — all present on Base per Section 7.2) itself a wrapped/bridged/restaked claim on a further off-chain or cross-chain process, and if so, what is that process's own historical incident record (Section 10.1 for the general pattern; a full per-asset dependency audit was not completed in this pass — see Section 16)	This is exactly the mechanism that produced the \$123.7–230.1M Aave exposure in the Kelp DAO incident; a materially complete risk methodology should not treat "listed on Aave" as sufficient information about a wrapped/restaked asset's own upstream dependency risk
GHO peg deviation and facilitator bucket utilization	GHO/USD price history (via its confirmed distinct oracle-source address, Section 7.2, once code-verified per Section 8.2) plus <code>getFacilitatorBucket()</code> calls per	GHO's peg mechanism is fundamentally different from a reserve-backed stablecoin's (Section 11.3); a peg-deviation signal for GHO is closer to

Candidate observation	Primary evidence required	Interpretation limits
GENIUS Act implementation-status milestones (NPRM comment-period closure, final-rule publication, first issuer approval) as a binary/staged regulatory-clarity indicator	facilitator (per aave.com/docs/ecosystem/gho) Direct monitoring of Federal Register dockets identified in Section 4.2	"Aave-protocol-collateral-quality signal" than to "third-party issuer reserve-adequacy signal," and should not be scored using the same rubric as USDC/EURC Regulatory clarity is not the same as reduced credit/liquidity/oracle risk for any specific instrument (the Foundation Document's own rule, restated: do not equate compliance with low risk); this signal, if used at all, should be scoped narrowly to "legal/operational clarity," not blended into a general safety score

15. Signals that should NOT be used, or that require extreme caution

Extends the Foundation Document's own table (its Section F) with new items from this pass's research.

Unsupported or hazardous shortcut	Reason
"GENIUS Act compliant" or "GENIUS Act approved" as a label applied to any current stablecoin (including any Aave Base reserve)	No final rule and no approved issuer exists as of 11 September 2026 (Section 4.2); the label is definitionally unavailable right now, so any such claim is either forward-looking marketing or an error
A Chainlink/Aave-address-book "oracle source" address matching between two assets, used as an automatic "these assets are equally safe" inference	Section 8.6 shows this occurs for USDC/USDbC (economically near-identical, low incremental risk from the sharing itself) and for cbBTC/tBTC (economically and custodially distinct, where sharing could mask a real divergence) — the same raw fact (shared address) supports opposite interpretations depending on the underlying assets' actual redemption mechanics, so it cannot be used as a standalone signal without the asset-specific context in Section 8.3
A cached/scraped current LTV, liquidation threshold, liquidation bonus, cap, debt ceiling, or reserve factor figure from any secondary aggregator or from this document itself, used without a same-day live on-chain re-verification	Section 5.5 confirms an automated Risk Steward/Edge Risk Oracle layer can change several of these parameters within governance-set bounds without a discrete announcement; any cached figure has an unknown and non-trivial staleness risk baked in
Audit count or audit-firm reputation (Section 10.2's list) as a safety proxy, independent of what specific version was audited and whether that version is the one actually deployed	The Kelp DAO/rsETH incident (Section 10.1) produced \$123–230M of Aave exposure with zero code vulnerability found in either Kelp DAO's or Aave's contracts — it was a governance/risk-parameter and cross-protocol-dependency failure, not an audit-detectable code bug, which is direct empirical evidence (not merely a theoretical caution) that audit depth does not bound this class of loss
"Base had X days without an outage" as a chain-reliability safety signal	Section 10.4 shows Base's own outage cadence and severity have varied considerably and non-monotonically over 2024–2026 (short stalls, a mis-signaled activity collapse, a genuine settlement-layer delay, and a genuine multi-incident block-production halt); a clean recent streak does not characterize the underlying single-sequencer

Unsupported or hazardous shortcut	Reason
Total Base TVL, or Base's rank among L2s by TVL secured, as a proxy for the safety of any specific Aave Base reserve or of Base as a settlement layer	architectural risk, which is structural and did not change across any of these incidents
GHO's peg holding at or near \$1.00 as evidence of the same class of reserve adequacy that a peg-holding fiat-backed stablecoin would demonstrate	TVL measures capital attracted, not resilience; the Kelp DAO incident and the June 2026 sequencer outages both occurred on/affected large-TVL networks (Ethereum, Arbitrum, Base) without TVL size preventing either Per Section 11.3, GHO's peg mechanism is an interest-rate/DAO-governance mechanism acting on a fully on-chain, over-collateralized borrow-mint system, not third-party reserve custody; a stable GHO price is evidence about Aave's own collateral/liquidation mechanics functioning, not about any external reserve custodian
A single secondary source's dollar or ETH figure for the Kelp DAO/rsETH incident's total loss, bad debt, or recovery-fund size, treated as precise and final	Multiple reputable outlets reported different, evolving figures across late April/May 2026 (e.g., \$292M vs. \$293M stolen; \$123.7M vs. \$230.1M bad-debt range; total DeFi United commitments reported at \$161M, \$314M, and other intermediate figures depending on the reporting date) because the situation was actively resolving; treat any single number as time-stamped to its source, not as the final outcome, absent a confirmed closing figure (which this pack does not have — Section 16)
Treating the Base-deployed Risk Steward/Edge Risk Oracle automation contracts (Section 5.5, 7.1) as neutral infrastructure with no governance-trust assumption	These contracts hold governance-delegated authority to change real risk parameters within bounds; their own permission scope, upgrade keys, and the specific bound constraints (e.g., the "roughly 2x" cap-increase limit mentioned in Chaos Labs' own description) are themselves a form of protocol-governance risk surface and should be catalogued (via <code>ACL_MANAGER</code> role checks, Section 12) rather than assumed benign because they are framed as "automation"

16. Known unknowns / missing evidence

This section consolidates every explicit gap flagged throughout this document, so the next model does not need to re-derive them from prose.

Aave V3 Base — exact current parameters (all [LIVE-QUERY REQUIRED], Section 12 gives method):

- Current LTV, liquidation threshold, liquidation bonus, supply cap, borrow cap, debt ceiling, and reserve factor for all 15 listed reserves.
- Which of the 15 eMode categories are currently flagged "isolated" under the still-being-debated v3.7 policy (Section 5.3), and each category's exact LTV/LT/LB.
- The exact, resolved oracle-source address for WETH and cbETH (two conflicting candidate addresses found, Section 8.5) and confirmation of every other asset's oracle-source address against live state, not just address-book text.
- Whether each per-asset oracle source is a raw Chainlink proxy, an Aave `PriceCapAdapterBase`, an Aave `PriceCapAdapterStable`, or a composed feed (Section 8.2) — determined for **zero** of the 15 assets in this pass.
- Whether the shared cbBTC/tBTC oracle address (Section 8.6) reflects a flat BTC/USD feed applied to both, or a more nuanced treatment — this is flagged as the single highest-priority follow-up item.
- Heartbeat, deviation threshold, and decimals for whichever underlying Chainlink feed each resolved adapter ultimately reads.
- The exact interest-rate-strategy parameters (optimal utilization, slopes, base rate) actually configured for each of the reserves sharing the common strategy contract address.
- The precise execution date/transaction of Base's v3.7 upgrade ("Part 2"), not just the secondary-sourced 29 May 2026 approximate date.
- The exact function/permission scope of the `CLINIC_STEWARD` and `POOL_EXPOSURE_STEWARD` contracts (only inferred from naming convention in this pass).
- Which reserves are currently frozen/paused and why (Section 7.5 noted several appear disabled for borrowing in a cached snapshot, cause unconfirmed).
- The content of the recent (secondary-press-reported, ~1 week before this review) Base parameter-update proposal (Section 9.1's last row).

Governance/history:

- A complete, exhaustive (not representative) list of every Base-specific asset listing, delisting, freeze, cap change, and oracle change since Base's original 2023 deployment.
- Whether Aave holds any historical bad-debt/deficit events specific to Base (none were found in this pass, but a targeted historical-incident search specific to Base was not exhaustively performed beyond the Kelp DAO cross-reference).

Security/incidents:

- The final, closed-out resolution of the Kelp DAO/rsETH incident's Aave bad-debt allocation and DeFi United funding (last confirmed status: active, unresolved, as of early May 2026 evidence retrieved in this pass).
- Whether any Base-specific liquidation was affected by the June 2026 Base sequencer outages, in light of the v3.7 sequencer-sentinel removal (Section 10.5) — not investigated at the transaction level in this pass.
- A dedicated pre-2026 Aave historical-incident catalogue (Safety Module activations, any pre-2026 bad debt, the 2022 UST freeze and its parameter-change specifics) was not compiled.
- Confirmation of Aave's "Umbrella" backstop mechanism directly from Aave's own documentation (only secondary-sourced in this pass, Section 5.4).
- Independent verification of Aave's bug-bounty program terms and the \$250K figure cited by a secondary source (Section 10.3).

RWA/stablecoin:

- Federal Reserve Board's GENIUS Act rulemaking status (not identified at all in this pass — an outright gap, not merely "proposed but not final").
- NCUA's rule, beyond a general secondary characterization.
- A complete state-by-state substantial-similarity/threshold-transition register.
- Circle's current, dated USDC/EURC reserve composition, attestation firm, and attestation date (not re-fetched in this pass; only the general caution about instrument-level evidence was carried forward).
- Whether Base has a local GHO facilitator (mint capability) or purely hosts a borrow/supply market for bridged-in GHO, and if the latter, the specific cross-chain bridging/GSM architecture used (Section 11.3).
- Full issuer-specific dossiers (the 8-item checklist in Section 11.4) for USDC, EURC, USDbC, and GHO as actually supplied/deployed on Base.

Data infrastructure:

- No live query of any indexer (Graph, Dune, or a paid Aavescan/API-key-gated endpoint) was actually executed in this pass; all "available data source" claims in Section 13 are about the existence and apparent scope of these sources, not about data pulled from them beyond what their own public marketing/documentation pages disclosed.

x402/ERC-8004/GENIUS-statute-level:

- No new research was conducted on these topics in this pass; all Foundation-Document-level gaps in these areas (Section "Important missing material" of the Foundation Document) remain exactly as that document left them.

17. Source registry

Legend: **P** = primary source (official documentation, enacted law, deployed/verified code, official governance record, first-party blog/press release). **S** = secondary source (journalism, community aggregators, unofficial analysis). Foundation-Document sources [^F1]–[^F30] are listed by their Foundation Document footnote number and are not re-numbered; see the Foundation Document itself for their full citation detail.

#	Title	Issuing organization		URL
	[^F1]- (carried forward in [^F30] full)	—	—	
N1	<code>AaveV3Base.sol</code>	Aave DAO (<code>aave-dao/aave-address-book</code>)		github.com/aave-dao/aave-address-book/blob/main/src/AaveV3Base.sol
N2	<code>AaveV3Base.sol</code> / <code>AaveV3Base.ts</code> (bgd-labs mirror)	bgd-labs		github.com/bgd-labs/aave-address-book
N3	Aave protocol Changelog	Aave		aave.com/docs/resources/changelog
N4	"[ARFC] BGD. Aave v3.7"	BGD Labs / Aave Governance forum		governance.aave.com/t/arfc-bgd-aave-v3-7/24075

#	Title	Issuing organization	URL
N5	"Upgrade Aave instances to v3.7 Part 1"	Aave Governance (execution record)	vote.onaave.com / governance.aave.com
N6	"[ARFC] Unified Handling of the Isolated Flag in Aave V3.7 E-Modes"	Aave Governance forum	governance.aave.com/t/arfc-unified-handling-of-the-isolated-flag-in-aave-v3-7-e-modes/25621
N7	aave-v3-origin/audits directory	Aave DAO	github.com/aave-dao/aave-v3-origin/tree/main/audits
N8	Aave Security page	Aave	aave.com/security
N9	Aavescan Base V3 market page and asset sub-pages	Aavescan	aavescan.com/base-v3 (and /usdc, etc.)
N10	Aave app markets page (Base)	Aave	app.aave.com/markets/?marketName=proto_base_v3
N11	"CBETH / ETH," "WSTETH / USD," "ETH / USD," "USDC / USD," "BTC / USD," "USDT / USD," "LINK / USD" Chainlink feed pages	Chainlink	data.chain.link/feeds/... (various, Ethereum/Optimism mainnet examples)

#	Title	Issuing organization	URL
N12	Base chain oracle-provider list	Base (Coinbase)	docs.base.org/chain/oracles
N13	aave-price-feeds repository README	Aave DAO	github.com/aave-dao/aave-price-feeds
N14	"[Direct to AIP] LBTC Oracle and CAPO implementation Update"	Chaos Labs / Aave Governance forum	governance.aave.com/t/direct-to-aip-lbtc-oracle-and-capo-implementation-update/22614
N15	"Aave could face up to \$230m in losses after Kelp DAO bridge exploit triggers DeFi chaos"	CoinDesk	coindesk.com, 20 Apr 2026
N16	"Kelp DAO Loses \$293M in Bridge Exploit, Leaving Aave With Over \$200M in Bad Debt"	The Defiant	thedefiant.io
N17	"Aave lays out rsETH risk and recovery paths after Kelp DAO exploit"	CryptoBriefing	cryptobriefing.com
N18	"AAVE wETH Exploit: \$200M Bad Debt Hits Depositors"	Forbes	forbes.com

#	Title	Issuing organization	URL
N19	"[ARFC] rsETH Incident Funding Update"	TokenLogic / Aave Governance forum	governance.aave.com/t/arfc-rseth-incident-funding-update/24740
N20	"[TEMP CHECK] Post-rsETH Collateral Framework: Tier-Based LTV Reductions and Wrap-Depth Ineligibility Limits"	Aave Governance forum (Robby Greenfield IV / Tokédex)	governance.aave.com/t/temp-check-post-rseth-collateral-framework-tier-based-ltv-reductions-and-wrap-depth-ineligibility-limits/24726
N21	"The Butterfly Effect Comes to DeFi"	NYDIG Research	nydig.com/research
N22	"Mantle's 30,000 ETH loan for Aave enters vote as DeFi United tops \$314m"	crypto.news	crypto.news
N23	"DeFi lost \$13B this month as the KelpDAO rescue shows both the best and worst of DeFi"	CryptoSlate	cryptoslate.com
N24	"Postmortem: June 25th Block Production Outage"	Base (Coinbase) engineering blog	blog.base.dev/postmortem-june-25th-block-production-outage
N25	"One Sequencer Bug Froze Base Twice in	CoinPaprika	coinpaprika.com

#	Title	Issuing organization	URL
	Two Days"		
N26	"Post Mortem: What Happened on Base on June 25, 2026"	Metrika	metrika.co/blog
N27	"Base Ethereum Withdrawals Stall 36 Hours After Azul Upgrade Bug"	CoinPaprika	coinpaprika.com
N28	"Base's state update system went down and nobody noticed"	Cryptopolitan	cryptopolitan.com
N29	Federal Register Doc. 2026-16796 (Treasury Section 3 NPRM)	U.S. Department of the Treasury	federalregister.gov/documents/2026/08/18/2026-16796
N30	OCC Bulletin 2026-3 / NPRM	Office of the Comptroller of the Currency	occ.gov/news-issuances/bulletins/2026/bulletin-2026-3.html ; federalregister.gov/documents/2026/03/02/2026-04089

#	Title	Issuing organization	URL
N31	Federal Register (FDIC NPRM, RIN 3064-AG19)	Federal Deposit Insurance Corporation	federalregister.gov/documents/2026/04/10/2026-06974
N32	Treasury press release (Section 3 NPRM announcement)	U.S. Department of the Treasury	home.treasury.gov/news/press-releases/sb0605
N33	Treasury press release (BSA/AML/sanctions NPRM)	U.S. Department of the Treasury	home.treasury.gov/news/press-releases/sb0435
N34	"GENIUS Act Pushes Stablecoin Compliance Into Banks' Back Offices"	PYMNTS	pymnts.com
N35–N39	Law-firm client alerts summarizing the OCC/FDIC/Treasury proposals (Morgan Lewis, Sullivan & Cromwell, K&L Gates, Gibson Dunn, Mayer Brown, Troutman)	Respective law firms	morganlewis.com ; sullcrom.com ; klgates.com ; gibsondunn.com ; mayerbrown.com ; troutmanfinancialservices.com
N40	GHO documentation	Aave	aave.com/docs/ecosystem/gho
N41	GHO Facilitators help page	Aave	aave.com/help/gho-stablecoin/facilitators
N42	aave-dao/aave-v3-risk-stewards repository	Aave DAO	github.com/aave-dao/aave-v3-risk-stewards

#	Title	Issuing organization	URL
N43	"Introducing Aave's Chaos Labs Risk Oracles"	Chaos Labs	chaoslabs.xyz/posts/introducing-aave-chaos-labs-risk-oracles
N44	"BGD. Risk Steward Phase 1: CapsPlusRiskSteward" / "Phase 2: RiskSteward"	BGD Labs / Aave Governance forum	governance.aave.com/t/bgd-risk-steward-phase-1-capsplusrisksteward/12602; .../arfc-bgd-risk-steward-phase-2-risksteward/16204
N45	<u>aave-dao/aave-proposals-reports</u> repository	Aave DAO	github.com/aave-dao/aave-proposals-reports
N46	<u>PaulieB14/graph-aave-mcp</u> repository	Community (PaulieB14)	github.com/PaulieB14/graph-aave-mcp
N47	Dune query/dashboard pages (multiple)	Dune community authors	dune.com/queries/585720; dune.com/chaininsight/aave-liquidations-overview; dune.com/dcooper/aave-liquidations
N48	"Aave V3 Markets Scraper"	Apify (ParseForge, community)	apify.com/parseforge/aave-v3-markets-scraper

#	Title	Issuing organization	URL
N49	aave/risk-v3 "risk-parameters.md"	Aave (older/generic repository)	github.com/aave/risk-v3/blob/main/asset-risk/risk-parameters.md
N50	QuickNode "Aave V3 Liquidation Tracker" sample-app documentation	QuickNode	quicknode.com/sample-app-library/ethereum-aave-liquidation-tracker
N51	Aave Markets Data documentation	Aave	aave.com/docs/aave-v3/markets/data

18. Consistency check (performed across this document before finalization)

- **Contract-address conflict identified and preserved (not silently resolved):** WETH and cbETH oracle-source addresses on Base differ between the [aave-dao/aave-address-book](#) direct-fetch result and a [bgd-labs/aave-address-book](#) cached-search-snippet result (Section 8.5). All other cross-checked addresses (POOL, POOL_CONFIGURATOR, ORACLE, ACL_MANAGER, POOL_EXPOSURE_STEWARD, and the full asset/eMode list) were consistent between every source consulted for them.

- **Date consistency:** the v3.7 "rollout completes" changelog entry, the "Part 1" execution list (which excludes Base), and the secondary-sourced 29 May 2026 completion date are mutually consistent (Base is absent from Part 1 and present in the "completes" list, with the secondary date falling after the Part-1-only networks would have been upgraded) but the exact Base execution date/transaction was not independently pinned — flagged in Section 16, not stated as more precise than it is.
- **Figure consistency:** the Kelp DAO incident's dollar/ETH figures vary across sources within plausible ranges consistent with a fast-evolving situation reported in real time (e.g., \$292M vs. \$293M; 116,500 rsETH is consistent across all sources checked). No source presented a figure that was irreconcilably contradictory (e.g., an order-of-magnitude difference) — the variation is treated as normal reporting-precision drift, not a red flag requiring a scoring caution beyond what Section 15 already states.
- **Version-number consistency:** every source touching Aave's protocol version for Base (changelog, ARFC, audit dates, secondary press) is internally consistent with "Base is on v3.7," with no source found in this pass claiming a different version for Base.
- **No conflicting figures were found and silently dropped** — every discrepancy identified during research is surfaced in Sections 7.5, 8.5, 9.1, or 15 rather than being resolved by silent selection of one value.

End of MASTER_RISK_ORACLE_EVIDENCE_PACK.md. This document, together with primary-source-knowledge-base.md, is intended as the complete evidentiary foundation handed to a separate reasoning model for risk-methodology design and red-teaming. That model should treat every item marked UNKNOWN / NOT VERIFIED or [LIVE-QUERY REQUIRED] as a precondition to resolve — via the exact contract/function calls in Section 12 and the exact follow-up items in Section 16 — before finalizing any scoring weight, threshold, or formula that depends on it.